Adapting 3-D CT Foundation Models Beyond Frozen Probes: What to Train, and One Model or Many

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Abstract

Pretrained 3-D CT foundation models are usually evaluated with a frozen encoder and a linear or attentive head. However, a group adapting such a model to new labels has to decide what to train, from a frozen probe through low-rank adapters to full fine-tuning, and whether to train one model per label or one for all labels of a cohort, and the evidence for these choices comes mostly from 2-D backbones and single tasks. In this work we compare 11 adaptation methods for multi-label classification and 7 for segmentation on four public 3-D CT encoders of three architectural families (TAP-CT, SPECTRE, Pillar-0 and VoCo) under one protocol: 39 binary labels from five public cohorts and 37 structures from five public datasets, each trained per task and jointly with one default setting and budget (1988 runs, plus seed repeats and controls), and compared at label level with intervals over cohorts, labels and held-out cases. For classification, adapting the attention weights of the encoder improves on both the linear and the attentive frozen probe on the three main backbones, and one joint model per cohort is as accurate as one model per label; on the hierarchical VoCo encoder, adapting the encoder was level with the matched attentive probe, and both were behind its linear probe. For segmentation, a multi-depth decoder on the frozen encoder gives most of the gain over a linear probe, adapting the encoder on top of it changes Dice little on average, and one model per structure is more accurate than one joint model at our budgets. Full fine-tuning with its default setting failed to train on one backbone and recovered with layer-wise learning-rate decay, and the LoRA variants did not improve on plain LoRA. We condense the results into a practical guide that separates consistent findings from conditional and uncertain ones, and release code, splits and per-case predictions.

Keywords: Foundation models, computed tomography, parameter-efficient fine-tuning, low-rank adaptation, multi-task learning, segmentation

1. Introduction

A group that wants to detect a new finding on CT, with a few hundred to a few thousand labeled scans, can now start from one of several publicly released 3-D CT encoders, among them TAP-CT (Veenboer et al., 2026), SPECTRE (Claessens et al., 2026), Pillar-0 (Agrawal et al., 2025), CT-FM (Pai et al., 2025), Merlin (Blankemeier et al., 2026), CT-CLIP (Hamamci et al., 2026), FM-HCT (Zhu et al., 2026) and VoCo (Wu et al., 2024, 2026); 2-D radiology models such as Curia process a volume slice by slice (Dancette et al., 2025). Their evaluations rank representations: TAP-CT reports a linear layer on frozen tokens for segmentation and an attention-pooled head on frozen class tokens for classification (Veenboer et al., 2026), SPECTRE a nearest-neighbour classifier on frozen embeddings

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(Claessens et al., 2026), and CT-FM end-to-end fine-tuning (Pai et al., 2025), while for FM-HCT probing and fine-tuning are compared on the same tasks (Zhu et al., 2026). However, such a table does not settle what the group has to decide before training: whether to keep the encoder frozen, adapt it with a low-rank update (Hu et al., 2022) or fine-tune part or all of it; whether to train one model per finding or one for all findings of a cohort; and, for segmentation, which head turns tokens into a voxel map. These choices change accuracy, compute and the number of models to maintain, and a frozen probe does not predict them: in our comparison, SPECTRE trails TAP-CT on the RSNA 2022 cervical-spine labels with the linear probe (-0.019 AUROC) but not once both are adapted with LoRA ($+0.002$), and full fine-tuning with a standard setting collapsed on one of the encoders.

The evidence for these choices comes mostly from 2-D backbones and single tasks. On 2-D medical classification, LoRA matches or surpasses full fine-tuning of ImageNet-pretrained transformers (Dutt et al., 2024; Zhu et al., 2024), and probing and fine-tuning of DINOv2 (Oquab et al., 2024) have been compared on radiology tasks (Baharoon et al., 2023). For 3-D classification, Liu et al. (2026b) found that frozen probing understates 2-D and native 3-D foundation models, and on one of our labels, RSNA 2023 spleen injury, full fine-tuning of a convolutional and a Swin 3-D model surpassed LoRA, which surpassed linear probing. Parameter-efficient adaptation of 3-D medical encoders has otherwise been studied mostly for segmentation: adapters that lift the 2-D Segment Anything model (Kirillov et al., 2023) to volumes (Gong et al., 2024; Chen et al., 2024a; Wu et al., 2025), prompt tuning of a self-supervised 3-D U-Net (Fischer et al., 2024), LoRA, adapters and full fine-tuning of a supervised 3-D CT model in few-shot transfer (Silva-Rodríguez et al., 2025), and automated adapter configuration for few-shot 3-D segmentation (Ly and Nguyen, 2026); for a 3-D CT vision–language model, full fine-tuning and parameter-efficient fine-tuning were close on one classification task (Kartika et al., 2026). Each of these studies uses one backbone and one task, and variants of LoRA that reparameterize or rescale the update (Liu et al., 2024; Kalajdzievski, 2023), probing before fine-tuning (Kumar et al., 2022) and fine-tuning only the attention weights (Touvron et al., 2022) were not part of them. Joint training on many labels can help (Kim et al., 2023) or hurt through conflicting gradients (Yu et al., 2020; Zhao et al., 2023), and task-specific LoRA weights generated by a hypernetwork have been proposed for chest CT (Liu et al., 2026a); the closest CT foundation-model studies do not compare joint and per-task adaptation of a 3-D encoder on the same labels (Veenboer et al., 2026; Liu et al., 2026b). For segmentation, nnU-Net (Isensee et al., 2021) and models pretrained on segmentation labels (Li et al., 2024; He et al., 2025) remain the references.

In this work, we compare what to train when adapting a pretrained 3-D CT encoder to new labels, under one protocol, on four public pretrained backbones of three architectural families: two plain vision transformers (TAP-CT, SPECTRE), a convolutional-stem multi-scale transformer (Pillar-0) and a hierarchical Swin transformer (VoCo). For classification we compare 11 methods, from frozen linear and attentive probes through low-rank adapters and their variants to attention-only and full fine-tuning, on 39 binary labels from five public cohorts. For segmentation we compare 7 methods, from a linear probe and a multi-depth decoder on the frozen encoder to adapters and full fine-tuning on top of the decoder, on 37 structures from five public datasets. Every method is trained once per label and once jointly for all labels of a cohort, with one default setting and budget, and compared at

label level with intervals over cohorts, labels and held-out cases. Our contributions can be summarized as follows:

- We provide a controlled comparison of 11 classification and 7 segmentation adaptation methods on four 3-D CT backbones (1988 training runs, plus 367 repeats and controls), trained per task and jointly, with per-case predictions released.
- We show which part of the encoder is worth training for each task and how far the answer transfers between backbone families, and we quantify the cost of one joint model against one model per label.
- We condense the results into a practical guide that separates consistent findings from conditional and uncertain ones.

2. Methods

Backbones. We adapt four publicly released pretrained 3-D CT encoders of three architectural families (Figure 2 in Appendix B). **TAP-CT-B-3D** (Veenboer et al., 2026), pretrained by DINOv2-style self-supervision, is a plain 12-block ViT-B (Dosovitskiy et al., 2021) with $4\times8\times8$ voxel patches, a class (CLS) token and four register tokens (Darcet et al., 2024), applied to windows of $12\times224\times224$ voxels at native slice spacing. **SPECTRE-Large** (Claessens et al., 2026), pretrained by self-distillation and vision–language alignment with radiology reports, is a plain 24-block ViT-L with $16\times16\times8$ patches on crops of $128\times128\times64$ voxels at $0.5\times0.5\times1$ mm, followed by a pretrained four-block feature combiner that attends over the crops of a volume. **Pillar-0** (Agrawal et al., 2025), pretrained on CT and breast MRI of one academic centre, is used through its released head-CT encoder, a convolutional stem followed by three stages of multi-scale window-attention blocks that encode a whole $128\times256\times256$ volume from eleven intensity windows in one pass; its pretraining on head CT restricts it to the hemorrhage cohort. **VoCo-L** (Wu et al., 2024, 2026) is the encoder of a SwinUNETR-v2 (Hatamizadeh et al., 2022a; He et al., 2023), a four-stage Swin transformer (Liu et al., 2021) with patch merging between stages and no class token, applied to 96^3 windows at 1.5 mm; we use the released self-supervised checkpoint. Volumes are reoriented, resampled and intensity-normalised as in each backbone's pretraining (Appendix B).

Classification heads and methods. Every backbone exposes a set of tokens per volume: the CLS token of every window (TAP-CT), the combiner tokens of the crops (SPECTRE), the multi-scale tokens of the whole volume (Pillar-0), or one token per window formed by the spatial means of the outputs of stages 2–4 (VoCo, 2688-dimensional). **Lin** reads a pooled feature (the mean window token, the combiner's CLS token or Pillar-0's pooled embedding) through dropout and one linear layer. **Attn** is an attentive probe (Bardes et al., 2024; Chen et al., 2024b): a learnable CLS token joins the token set, two self-attention blocks run over it, and the CLS output is read out. The 11 methods differ only in what is trained inside the encoder (Figure 1): the *frozen probes* Lin and Attn; **LoRA** (Hu et al., 2022) on the query, key and value projections of every block (rank 16, $\alpha = 32$, dropout 0.1) and its variants $\textbf{LoRA}_{\textbf{late}}$ (the last third of the blocks of TAP-CT and SPECTRE, the last stage of Pillar-0 and VoCo), **LoRA+ABMIL** (tokens pooled by gated attention (Ilse et al., 2018) before the head), **DoRA** (Liu et al., 2024) and **rsLoRA** (Kalajdzievski, 2023) (update scaled by

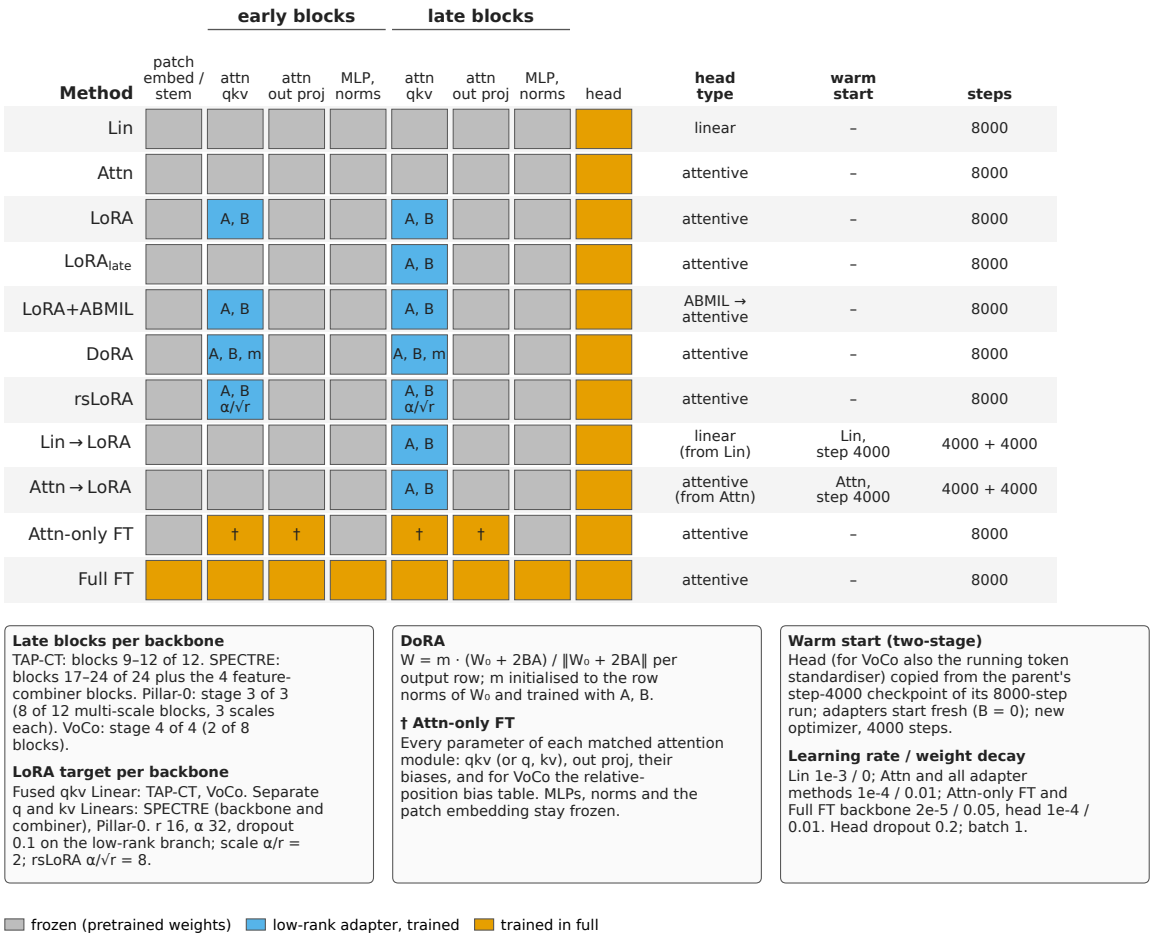


Figure 1: Overview of the classification methods, drawn from the implementation: what each method trains inside every encoder block and the head (grey = frozen, blue = low-rank adapter, orange = trained in full), the head it uses, its warm start and its budget, with the late blocks and adapted projections per backbone. The data path of every backbone is drawn in Figure 2 and the segmentation methods in Figure 4 (Appendix B).

$\alpha/\sqrt{r}$, four times the LoRA scale); the *two-stage schedules* **Lin→LoRA** and **Attn→LoRA**, which train the probe for half the budget and warm-start late-block LoRA from it, in the spirit of probing before fine-tuning (Kumar et al., 2022); and **Attn-only FT**, which trains every parameter of the attention modules (Touvron et al., 2022), and **Full FT**, which trains every parameter. All adapted and fine-tuned models use the attentive head; Lin→LoRA keeps the linear head of its parent.

Segmentation heads and methods. The encoder yields a grid of tokens per crop from four depths: TAP-CT blocks 3, 6, 9 and 12 (one token per $4\times8\times8$ voxels), SPECTRE blocks 6, 12, 18 and 24 (per $8\times16\times16$ voxels) and the outputs of VoCo's four stages (strides 4 to

32). **Lin** is a $1 \times 1 \times 1$ convolution on the deepest grid, upsampled trilinearly. **Dec** is a light multi-depth decoder in the spirit of UNETR (Hatamizadeh et al., 2022b): the four grids are projected, concatenated, fused by a 3^3 convolution and upsampled to the voxel grid in stages (about 2.5 M parameters, no skip connections from the input; Figure 3). **LoRA**, **LoRA$_{\text{late}}$** and **DoRA** add the adapters of classification under the Dec head, **Full FT** trains the encoder with it, and **Dec→LoRA** trains Dec for half the budget and warm-starts all-block LoRA from it (Figure 4). Predictions are independent sigmoids per structure.

Per-task and joint training. Every cohort and dataset is trained in two modes with the same data and selection rule (Figure 5): *per-task*, one model per label or structure, and *joint*, one shared model with one output per label or structure. A *label–mode pair* denotes one label (or structure) on one backbone in one mode. The joint classification loss sums over the labels known for a case, so a label weighs the same in both modes; the joint segmentation loss averages Dice and cross-entropy over structures. A joint model trains for the budget of one per-task model, so it uses $1/K$ of the steps of the K per-task models it replaces, except where stated below.

3. Experimental setup

Data. Classification uses five public CT cohorts with 39 binary labels (Table 2): the CVPR26 cohort of the CVPR 2026 3-D CT foundation-model challenge, which groups AMOS (Ji et al., 2022) with 15 disease labels, COVID-CTset (Rahimzadeh et al., 2021) and LUNA25 nodule malignancy (Peeters et al., 2026); FDG-PET/CT (Gatidis et al., 2022), whose four mutually exclusive diagnosis groups we predict from the CT alone as one-vs-rest labels; RSNA 2022 cervical-spine fractures (Lin et al., 2023) (patient level and C1–C7); RSNA 2023 abdominal trauma (Rudie et al., 2024) (any injury and kidney, liver and spleen injury); and RSNA 2019 intracranial hemorrhage (Flanders et al., 2020) (any hemorrhage and five subtypes). Segmentation uses AMOS22 (15 organs; Ji et al., 2022), KiTS23[1] (kidney, tumor, cyst; Heller et al., 2023), LiTS (liver, tumor; Bilic et al., 2023), MSD Pancreas (pancreas, tumor; Antonelli et al., 2022) and TotalSegmentator (Wasserthal et al., 2023; Wasserthal, 2026), whose 117 structures we merge within anatomical systems into 15 (Appendix A); 37 structures in all. Every source is divided into a training, a selection and a held-out split: for the CVPR26 cohort and AMOS22 the official validation cases are divided in halves, for the other sources the splits are fixed seeded divisions of the released labelled cases, with the official TotalSegmentator test split held out. Patients do not span splits where patient identifiers exist (FDG-PET/CT, RSNA 2023, RSNA 2019); the other sources are split per case. Pillar-0 is run on the hemorrhage cohort, and VoCo on FDG-PET/CT, RSNA 2022 and RSNA 2023 and on all segmentation datasets.

Training. Every run uses AdamW with a linear warm-up over the first 10% of the steps, cosine decay to 10^{-6}, bf16 mixed precision and seed 42. A classification run trains 8000 steps at batch size 1 on a masked, positive-weighted binary cross-entropy; a segmentation run trains 8000 steps of four crops on soft Dice plus binary cross-entropy per structure; two-stage methods split the budget into halves. The joint models of the two 15-structure datasets (AMOS22, TotalSegmentator) train 24 000 steps, and the CVPR26 joint model is

1. https://github.com/neheller/kits23

additionally trained for 26 000 steps as a longer schedule reported in the appendix. Each method has one default learning rate and weight decay, set once before the sweep by the type of parameter it trains, from the largest rate for a head on frozen features (10^{-3}; 10^{-2} for the segmentation linear probe) through low-rank adapters (10^{-4}) to the smallest for encoder weights (2×10^{-5}), and not tuned on our data for any backbone, cohort or method (Tables 4 and 5). Every 1000 steps both the selection and the held-out split are evaluated; the checkpoint with the best selection macro AUROC or mean Dice is kept and its held-out metric is reported. TotalSegmentator per-structure models use Lin, Dec, LoRA and Full FT only. The joint models of Lin, LoRA, Attn-only FT and Full FT (classification) and of Dec, LoRA and Full FT (segmentation) were trained with two further seeds on TAP-CT, SPECTRE and Pillar-0 (72 and 60 runs). All settings are listed in Appendix B.

Evaluation and statistics. Held-out AUROC per label and Dice per structure and case are the metrics (Maier-Hein et al., 2024). Per label–mode pair, methods are compared with DeLong's test on AUROC (DeLong et al., 1988) and with a paired case bootstrap on Dice, with Benjamini–Hochberg correction (Benjamini and Hochberg, 1995) within each family of comparisons at $\alpha = 0.05$, and ranked within each pair that has every method. The main comparisons are summarised at label level: each label counts once with its per-task and joint differences averaged, and the 95% interval of the mean comes from a percentile bootstrap over cohorts, labels within cohorts and held-out cases within labels (Appendix C); because four or five cohorts per backbone make the cohort stage optimistic, we also report the number of cohorts with a positive mean difference. Pooled estimates use TAP-CT, SPECTRE and Pillar-0; VoCo, which ran three of the five classification cohorts, is reported per backbone. A method whose held-out probabilities barely vary between cases did not train: we flag it when the median standard deviation of its held-out probabilities over label–mode pairs is below 0.01, keep it in the per-cohort tables and leave it out of ranks and averages over methods.

4. Results

We report held-out results of 1988 main runs and 367 repeats and controls. Table 1 summarises the main comparisons (per backbone in Figure 10, Appendix F); per-cohort tables, ranks, per-label results and every paired test are in Appendix F. The selection-split metric recomputed from the stored per-case predictions matched the value logged during training for every run (classification to 10^{-7}; segmentation within 2×10^{-3} for all but one of the 1034 runs, whose per-case Dice was re-inferred after training).

4.1. Classification

Adapting the encoder improves on the frozen probes. LoRA raises held-out AUROC over the linear probe by +0.034 at label level (95% interval +0.017 to +0.052; positive in 9 of 10 cohort–backbone pairs), with intervals above zero on each backbone: +0.032 on TAP-CT, +0.037 on SPECTRE and +0.033 on Pillar-0. LoRA also changes the head, from linear to attentive, but the frozen attentive probe alone adds only +0.007 (−0.012 to +0.023), and against this matched-head baseline LoRA gains +0.027 (+0.012 to +0.048; positive in 10 of 10 cohort–backbone pairs), so the gain comes from adapting the encoder.

Table 1: Label-level comparisons on the three main backbones: mean difference in held-out AUROC (classification) or Dice (segmentation), its 95% interval over cohorts, labels and held-out cases, and the number of cohorts (datasets) with a positive mean difference. Each label counts once with its per-task and joint differences averaged, except in the two LoRA − Dec sub-rows. LoRA − Attn compares two methods with the same attentive head. Pooled estimates ("all") use TAP-CT, SPECTRE and Pillar-0; VoCo is in Table 12. † = did not train with the default setting (Section 4.1).

Comparison	TAP-CT	SPECTRE	Pillar-0	All	Cohorts > 0
Classification (AUROC)					
LoRA − Lin	+0.032 [+0.007, +0.053]	+0.037 [+0.009, +0.075]	+0.033 [+0.021, +0.044]	+0.034 [+0.017, +0.052]	9/10
LoRA − Attn	+0.022 [+0.006, +0.042]	+0.033 [+0.003, +0.078]	+0.034 [+0.021, +0.052]	+0.027 [+0.012, +0.048]	10/10
Attn-only FT − LoRA	+0.019 [-0.001, +0.054]	+0.002 [-0.037, +0.026]	+0.001 [-0.002, +0.004]	+0.011 [-0.007, +0.029]	7/10
Full FT − Lin	+0.022 [-0.061, +0.068]	-0.117 † [-0.195, -0.059]	+0.004 [-0.013, +0.021]	-0.038 [-0.103, +0.019]	5/10
joint − per-task	+0.012 [-0.014, +0.038]	-0.003 [-0.044, +0.051]	-0.008 [-0.013, -0.001]	+0.004 [-0.021, +0.030]	6/10
Segmentation (Dice)					
Dec − Lin	+0.085 [+0.055, +0.109]	+0.061 [+0.027, +0.088]	–	+0.073 [+0.049, +0.092]	10/10
LoRA − Dec	-0.005 [-0.021, +0.024]	+0.023 [-0.018, +0.076]	–	+0.009 [-0.015, +0.046]	8/10
LoRA − Dec, per-task	-0.021 [-0.036, +0.002]	+0.006 [-0.003, +0.031]	–	-0.007 [-0.022, +0.009]	7/10
LoRA − Dec, joint	+0.011 [-0.021, +0.066]	+0.040 [-0.039, +0.147]	–	+0.026 [-0.021, +0.093]	8/10
Full FT − Dec	0.000 [-0.024, +0.016]	-0.003 [-0.035, +0.009]	–	-0.002 [-0.018, +0.009]	6/10
joint − per-task	-0.046 [-0.090, -0.008]	-0.058 [-0.093, -0.009]	–	-0.052 [-0.082, -0.022]	0/10

Leaving out FDG-PET/CT and COVID-CTset, two sources in which we did not rule out shortcuts, leaves it unchanged (+0.037, +0.019 to +0.057). Per label–mode pair, LoRA is never significantly behind the linear probe and significantly ahead in 17%, 9% and 83% of the pairs on TAP-CT, SPECTRE and Pillar-0; significance follows the held-out size rather than the effect (mean gain +0.034 and 0% significant below 100 held-out cases, +0.031 and 88% above 400; Figure 13). LoRA is ahead of the linear probe in 27 of 30 cohort–seed pairs of the seed repeats, and the gain is largest where the probe is weakest (+0.054 for labels whose probe reaches a selection AUROC of at most 0.6, +0.025 above 0.8).

Training the attention weights in full has the best mean rank, but its gain over LoRA is uncertain. At the default settings and over the 156 label–mode pairs, attention-only fine-tuning has the best mean rank (3.91; LoRA 4.76, LoRA+ABMIL 4.84, DoRA 5.03, linear probe 7.55; Figure 11). Its label-level difference to LoRA is +0.011

(−0.007 to +0.029): +0.019 on TAP-CT but +0.002 on SPECTRE and +0.001 on Pillar-0. The LoRA variants do not improve on LoRA: on the same pairs, $\text{LoRA}_{\text{late}}$ changes AUROC by −0.007, DoRA by −0.002, rsLoRA by −0.018, LoRA+ABMIL by +0.002, and the two-stage schedules by −0.017 (Lin→LoRA) and −0.022 (Attn→LoRA). The rsLoRA deficit on TAP-CT is a learning-rate effect: at a quarter of the rate, which matches its update scale to LoRA, it gains +0.029 (Appendix D).

Full fine-tuning did not train on SPECTRE with the default setting. The held-out probabilities of SPECTRE Full FT barely vary between cases (median standard deviation 0.004, against at least 0.114 for every other method on the three backbones), and it loses −0.117 to the linear probe. On TAP-CT it changes AUROC by +0.022 (−0.061 to +0.068), with one collapsed joint run (FDG-PET/CT, 0.608), and it is the least stable method across seeds (median between-seed change of the cohort mean 0.060, LoRA 0.008). Of three alternative settings, layer-wise learning-rate decay (0.75 per block) gave SPECTRE Full FT the highest selection AUROC (0.766, against 0.720 with a lower learning rate and 0.575 with gradient accumulation and clipping, which trained on 1 of 4 cohorts); with it Full FT is ahead of LoRA by +0.017 on SPECTRE and level with it on TAP-CT (+0.001; Table 8), a comparison that favours Full FT, since LoRA was not re-tuned.

One joint model per cohort is as accurate as one model per label. Joint minus per-task AUROC is +0.004 (−0.021 to +0.030), at $1/K$ of the training steps of the K per-task models; the sign varies by cohort from −0.035 to +0.049 (Figure 14), and on Pillar-0 the joint model is behind on average, with an interval below zero (−0.008, −0.013 to −0.001), though not for every method (Table 16).

4.2. Segmentation

A decoder on the frozen encoder gives most of the gain. Replacing the linear probe by the frozen decoder raises Dice by +0.073 (+0.049 to +0.092), on every dataset of both backbones and significantly on 93% (TAP-CT) and 86% (SPECTRE) of the structure–mode pairs. That is all of LoRA's gain over the linear probe on TAP-CT (+0.080) and most of it on SPECTRE (+0.085).

Adapting the encoder on top of the decoder changes Dice little on average. LoRA changes Dice relative to the frozen decoder by +0.009 (−0.015 to +0.046; −0.005 on TAP-CT, +0.023 on SPECTRE). The sign depends on mode and dataset (Figure 12): the largest gain is the SPECTRE AMOS22 joint model, where the decoder (0.658) falls below the linear probe (0.682) and LoRA reaches 0.784, while on the TotalSegmentator joint model LoRA loses on both backbones (0.733 against 0.770 on SPECTRE). LoRA is significantly ahead of the decoder on 57% of SPECTRE pairs and behind on 8%; on TAP-CT on 31% and 30%. Full fine-tuning has the best mean rank (2.94; LoRA 3.22, Dec 4.03), on the strength of TAP-CT, but its mean change relative to the decoder is −0.002 (−0.018 to +0.009) because 2 TAP-CT structure–mode pairs lose more than 0.1 Dice. DoRA, $\text{LoRA}_{\text{late}}$ and Dec→LoRA change Dice relative to the decoder by +0.020, +0.014 and −0.008, each with an interval that includes zero.

One model per structure is more accurate than one joint model at our budgets. One joint model per dataset changes Dice by −0.052 (−0.082 to −0.022), on none of the

10 dataset–backbone pairs positive, most on the 15-structure AMOS22 (−0.072, −0.076) and least on LiTS (−0.004, 0.000). Much of the cost comes from small structures the joint model loses entirely: in 28 of 428 structure–method pairs, all on the adrenal glands and the esophagus of AMOS22 and the merged glands of TotalSegmentator, the joint model scores below 0.01 Dice where the per-structure model reaches 0.407 to 0.784; without them the mean difference over structure–method pairs moves from −0.050 to −0.010 on TAP-CT and from −0.058 to −0.024 on SPECTRE. Tripling the steps of the 15-structure joint models gained +0.121 and +0.147, and at matched total compute (per-structure models of 1600 steps) the joint model is ahead with LoRA (+0.122 to +0.136 on AMOS22) but behind with the frozen decoder and full fine-tuning in three of the four dataset–backbone pairs (Table 9). The gain from tripling the steps shows that the segmentation budget was short; 75% of segmentation runs also selected one of their last two validations, against 49% in classification, a share that the cosine decay of the learning rate inflates and that therefore indicates rather than measures under-training. The segmentation comparisons rank methods under a fixed budget, not at convergence.

4.3. A fourth backbone: the hierarchical VoCo encoder

VoCo differs from the three main backbones in its token interface (no class token, one stage-pooled token per 96^3 window) and ran three of the five classification cohorts (16 labels) with the settings of the other backbones, so we report it separately (Appendix E). In segmentation the findings hold: the decoder adds +0.680 Dice to a linear probe on the stride-32 output that segments almost nothing, LoRA changes Dice relative to the decoder by −0.018 (−0.047 to +0.016), one joint model costs −0.041, and full fine-tuning is ahead of the decoder by +0.015 (+0.007 to +0.023). In classification the picture differs: LoRA changes AUROC relative to the linear probe by −0.071 (−0.168 to −0.005), behind on all 3 cohorts, but more than half of this deficit comes from the head: the frozen attentive probe is itself behind the linear probe (−0.040), and against this matched-head baseline every method that adapts the encoder is level, with intervals including zero (LoRA −0.031, attention-only fine-tuning +0.005, full fine-tuning +0.006; Figure 9). The only adapted method level with the linear probe is Lin→LoRA, which keeps the linear head (+0.011, −0.008 to +0.031). On VoCo, adapting the encoder therefore did not improve on a frozen probe with the same head, unlike on the three main backbones, and the attentive head on VoCo's 2688-dimensional window tokens was worse than a linear one. One joint model is again level with per-task models (−0.001). VoCo was run with the settings of the other backbones; whether a learning rate, head or window token chosen for it would change this we did not test, so the result bounds the transfer of the classification finding at the default settings rather than explaining it.

5. Discussion

A practical guide. Each step gives the expected change and the strength of the evidence: *consistent* (the label-level interval excludes zero with the same sign on TAP-CT and SPECTRE), *conditional* (the sign depends on the backbone, mode or budget) or *uncertain* (the interval includes zero).

1. **Classification: adapt the attention weights.** Rank-16 LoRA on the query, key and value projections of every block, with an attentive head, adds +0.034 AUROC over the linear probe (*consistent*); training the attention projections in full adds +0.019 over LoRA on TAP-CT only (*uncertain*). Compare against both frozen probes on selection data: on VoCo, with the same settings, adapting the encoder was level with the attentive probe and both were behind the linear probe (*conditional* on the backbone). These are default settings; only full fine-tuning was re-tuned.
2. **Classification: one joint model per cohort is a reasonable default.** It is level with per-task models on average (+0.004) at one run instead of one per label, with a sign that varies by cohort (*uncertain* for a given cohort).
3. **Segmentation: put a multi-depth decoder on the frozen encoder, and adapt the encoder only if it helps on selection data.** The decoder adds +0.085 and +0.061 Dice over a linear probe (*consistent*); LoRA on top of it changed Dice by −0.005 and +0.023 (*conditional*).
4. **Segmentation: train one model per structure when Dice matters most, or give a joint model more steps.** A joint model changed Dice by −0.046 and −0.058 at our budgets (*consistent*), and tripling its steps added +0.121 and +0.147; at matched compute the sign depends on the method (*conditional*).
5. **Treat full fine-tuning and the LoRA variants as options to validate.** With the default setting full fine-tuning did not train SPECTRE, so check that predictions vary between cases; layer-wise learning-rate decay made it train. In segmentation it can lose a structure the decoder segments, so monitor per-structure selection Dice (*conditional*). The LoRA variants were level with plain LoRA or below it (*uncertain*).

Limitations. Each method has one default learning rate and budget, set without tuning, and only full fine-tuning was retrained with other settings after it failed; full fine-tuning and rsLoRA proved sensitive to the default, so the ranking among the stronger methods may change with tuning. Only the joint models of seven method–task pairs were repeated with further seeds. Segmentation had not converged, and its results rank adaptation methods under one budget rather than measure how close these models come to task-specific networks; Dice is computed on each backbone's working grid without post-processing. SPECTRE's pretraining images include several of our evaluation sets, which affects comparisons between backbones (Appendix F). Pillar-0 enters through one cohort and VoCo through three. For most sources the splits are our own seeded divisions of the released labelled cases; we release them with the per-case predictions. One author co-developed TAP-CT; all backbones use the same protocol and settings.

Conclusion. Adapting the attention weights of a pretrained 3-D CT encoder improves classification over both frozen probes on TAP-CT, SPECTRE and Pillar-0, and one joint model per cohort is as accurate as one model per label; on the hierarchical VoCo encoder, at the same default settings, adapting the encoder was level with the matched frozen probe. For segmentation, a multi-depth decoder on the frozen encoder gives most of the gain, adapting the encoder on top of it changes Dice little on average, and one model per structure is more accurate than one joint model at our budgets. Code, splits and per-case predictions are released at [TODO: public repository URL].

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Appendix A. Data

Table 2: Cohorts and datasets. Labels = binary labels (classification) or structures (segmentation); counts are cases in the training / selection / held-out splits as read from the manifests. Split rule: "official val, halves" = the official validation cases sorted by identifier and alternately assigned to the selection and held-out split; "seeded" = one fixed seeded division of the released labelled cases (80/10/10 of the released training data for the three RSNA and FDG cohorts, 70/15/15 for KiTS23 and MSD Pancreas, 91/20/20 volumes for LiTS); TotalSegmentator holds out its official test split and takes an equally large selection split from the official training split. FDG-PET/CT cases are studies (1014 studies of 900 patients; no patient spans splits); RSNA 2019 training cases are a stratified subset of the released training patients. Backbones: T = TAP-CT, S = SPECTRE, P = Pillar-0, V = VoCo.

Cohort / dataset	Labels	Split rule	Backbones	Train	Sel.	Held-out
Classification						
CVPR26: AMOS	15 disease labels	official val, halves	T S	1463[†]	178	178
CVPR26: COVID-CTset	COVID-19	official val, halves	T S	308	32	32
CVPR26: LUNA25	nodule malignancy	official val, halves	T S	1000	100	100
FDG-PET/CT	4 diagnosis groups	seeded, patient level	T S V	776	116	122
RSNA 2022 C-spine	patient, C1–C7 (8)	seeded, per study	T S V	1615	202	202
RSNA 2023 trauma	any, kidney, liver, spleen (4)	seeded, one series per patient	T S V	2517	315	315
RSNA 2019 hemorrhage	any + 5 subtypes (6)	stratified, patient level	T S P	5000	3269	3264
Segmentation						
AMOS22	15 organs	official val, halves	T S V	200	50	50
KiTS23	kidney, tumor, cyst	seeded	T S V	343	73	73
LiTS	liver, tumor	seeded	T S V	91	20	20
MSD Pancreas	pancreas, tumor	seeded	T S V	197	42	42
TotalSegmentator	15 merged structures	official test held out	T S V	1721	109	109

[†] distinct AMOS cases over the 15 per-label files; each label is known for a subset of cases.

Table 3: TotalSegmentator: the 117 released structures merged into 15 within anatomical systems, as implemented. Where two source masks overlap, the voxel is assigned to the smaller structure.

Structure	Source structures
lungs and airway	five lung lobes, trachea
heart	heart, left atrial appendage
arteries	aorta, brachiocephalic trunk, left and right subclavian, common carotid and iliac arteries
veins	superior and inferior vena cava, left and right brachiocephalic and iliac veins, portal and splenic vein, pulmonary vein
liver	liver, gallbladder
spleen	spleen
pancreas	pancreas
kidneys	left and right kidney, left and right kidney cysts
GI tract	esophagus, stomach, duodenum, small bowel, colon
bladder and prostate	urinary bladder, prostate
glands	thyroid, left and right adrenal gland
brain and spinal cord	brain, spinal cord
spine	25 vertebrae (C1–L6), sacrum
other bones	24 ribs, sternum, costal cartilages, clavicles, scapulae, humeri, hip bones, femurs, skull
muscles	gluteus maximus, medius and minimus, iliopsoas, autochthon (left and right)

Appendix B. Training and evaluation details

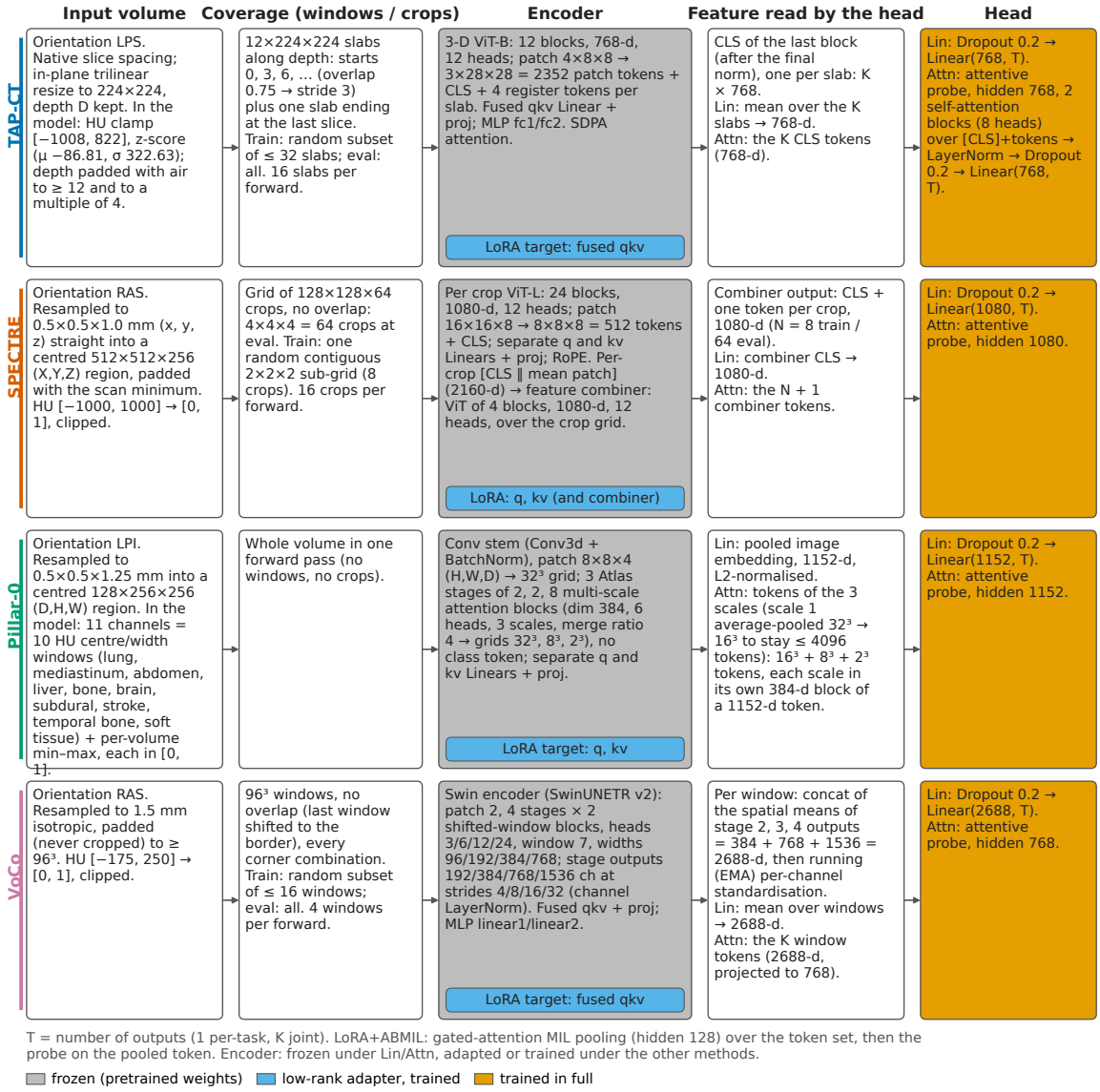


Figure 2: Overview of the classification path, drawn from the implementation. Each row is one backbone: the volume is resampled and windowed as in pretraining, covered by windows or crops, encoded, and reduced to the feature the head reads. Lin reads one pooled feature per volume, Attn the set of window, crop or scale tokens. Grey = frozen, blue = low-rank adapter, orange = trained in full.

Input preprocessing. Every volume is reoriented to the frame of the backbone's pretraining and resampled to its grid. TAP-CT: LPS, native slice spacing, in-plane trilinear

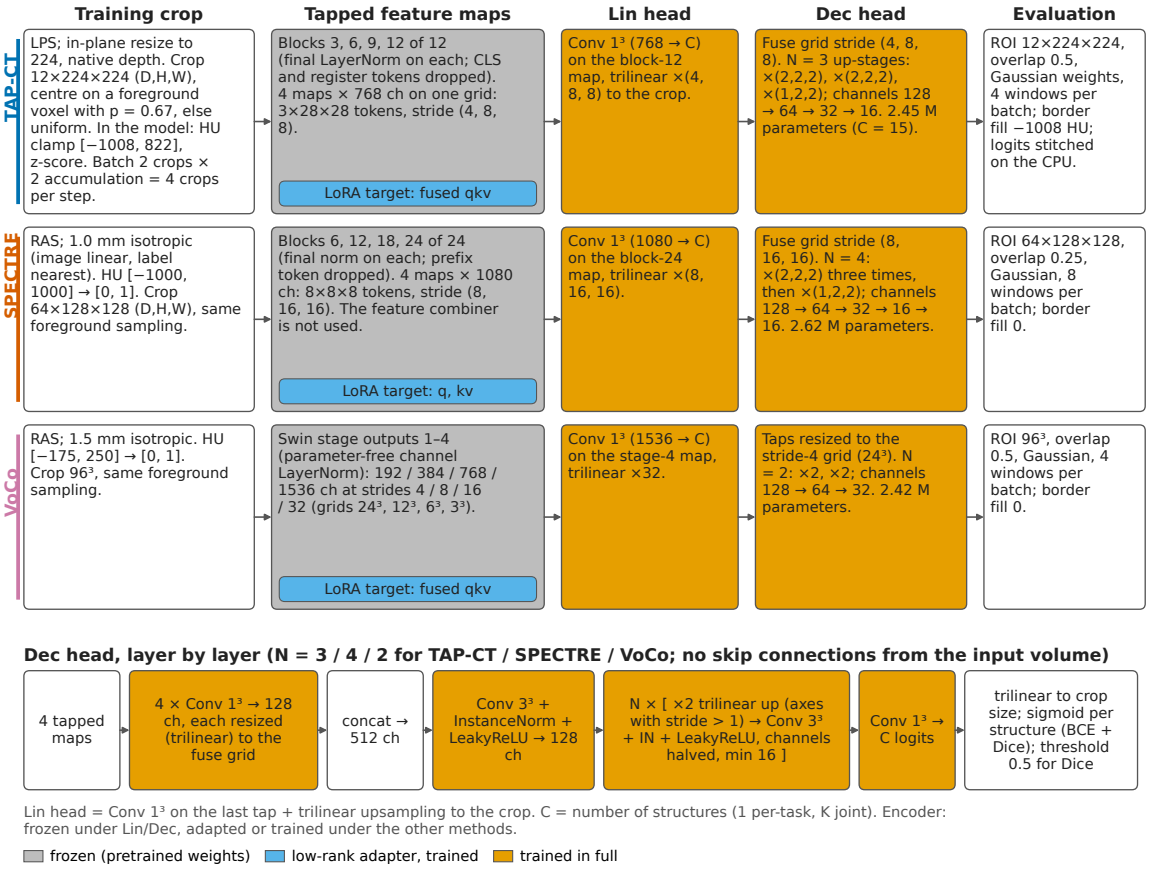


Figure 3: Overview of the segmentation path, drawn from the implementation: crop geometry per backbone, the four tapped depths with their strides and channels, the linear probe on the deepest grid, and the multi-depth decoder layer by layer. Grey = frozen, blue = low-rank adapter, orange = trained in full.

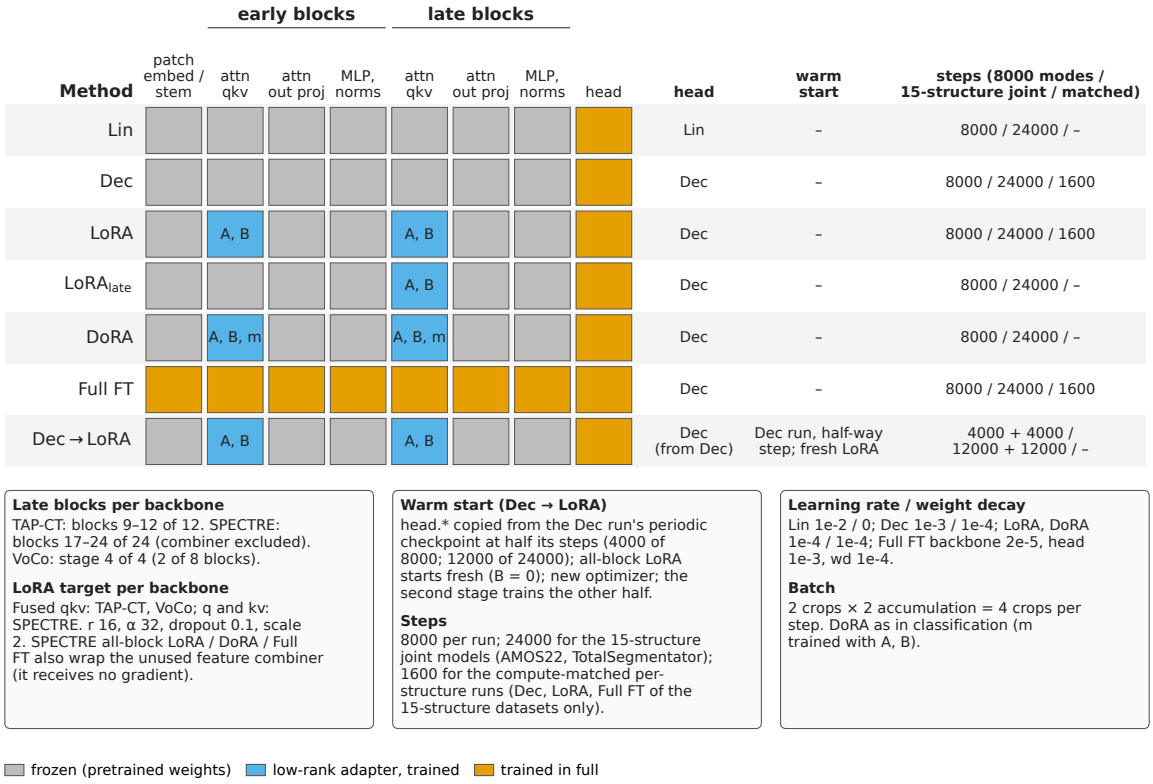


Figure 4: What each segmentation method trains, the head it uses, its warm start and its budget.

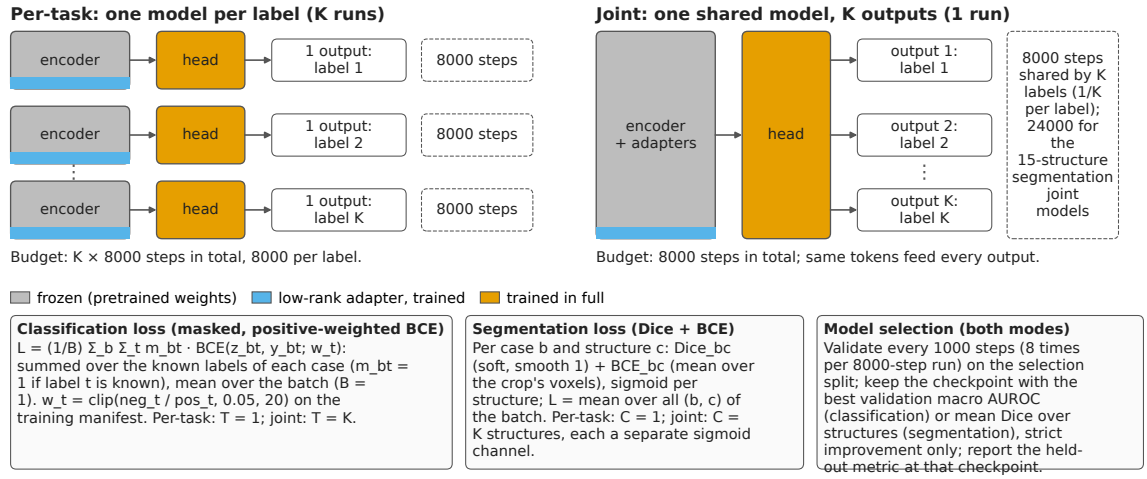


Figure 5: Per-task and joint training, the two losses, the step budgets and the selection rule.

resize to 224×224, depth padded with air to at least 12 slices and to a multiple of 4; HU clipped to $[-1008, 822]$ and standardised with the model's mean and standard deviation; windows of 12 slices with a stride of 3 slices plus a final window ending at the last slice (at most 32 windows per training step, all windows at evaluation). SPECTRE: RAS, $0.5 \times 0.5 \times 1$ mm, centred $512 \times 512 \times 256$ field of view padded with the scan minimum, HU in $[-1000, 1000]$ scaled to $[0, 1]$; a $4 \times 4 \times 4$ grid of 64 crops of $128 \times 128 \times 64$ voxels at evaluation and a random contiguous $2 \times 2 \times 2$ sub-grid of 8 crops at training; each crop is summarised by its CLS token concatenated with its mean patch token before the feature combiner. Pillar-0: LPI, $0.5 \times 0.5 \times 1.25$ mm, centred $128 \times 256 \times 256$ volume, eleven input channels (ten fixed centre/width windows and a per-volume min–max channel), one forward pass; the linear head reads the model's L2-normalised pooled embedding (three scales concatenated, 1152-d), the attentive head reads the tokens of the three scales, any scale with more than 4096 tokens being average-pooled on its grid. VoCo: RAS, 1.5 mm isotropic, HU in $[-175, 250]$ scaled to $[0, 1]$, padded to at least 96^3 and covered by non-overlapping 96^3 windows (the last window shifted to the border; at most 16 windows per training step); each window becomes one 2688-dimensional token, the concatenation of the spatial means of the stage-2, 3 and 4 outputs, standardised per channel with running statistics (momentum 0.01) that are stored with the checkpoint. For segmentation, TAP-CT keeps the classification grid (in-plane 224, native depth) and crops of $12 \times 224 \times 224$ voxels; SPECTRE resamples to 1 mm and uses crops of $64 \times 128 \times 128$ voxels; VoCo resamples to 1.5 mm and uses 96^3 crops. A training crop is centred on a random foreground voxel of the run's structures with probability 0.67 and placed uniformly otherwise; volumes smaller than the crop are padded with the image minimum. At evaluation the whole volume is covered by a sliding window with Gaussian weighting (overlap 0.5 for TAP-CT and VoCo, 0.25 for SPECTRE) and Dice is computed on the backbone's working grid, without post-processing.

Heads. The linear head is dropout (0.2) followed by one linear layer on the pooled feature: the mean of the window CLS tokens for TAP-CT (768-d), the feature combiner's CLS token for SPECTRE (1080-d), the pooled image embedding for Pillar-0 (1152-d) and the mean window token for VoCo (2688-d). The attentive probe prepends a learnable CLS token to the token set (the window CLS tokens for TAP-CT, the combiner tokens for SPECTRE, the multi-scale tokens for Pillar-0, the window tokens for VoCo; VoCo's tokens are first projected from 2688 to 768 dimensions), applies two pre-norm transformer blocks with eight-head self-attention and a GELU feed-forward layer of ratio 4, normalises the CLS output and maps it through dropout (0.2) and a linear layer. LoRA+ABMIL pools the window or crop tokens by gated attention (Ilse et al., 2018), $a_i = \mathrm{softmax}_i\big(\boldsymbol{w}^\top(\tanh(\boldsymbol{V}\boldsymbol{x}_i) \odot \sigma(\boldsymbol{U}\boldsymbol{x}_i))\big)$ with hidden size 128, and feeds the pooled token to the attentive probe. The segmentation linear probe is a $1 \times 1 \times 1$ convolution on the deepest grid (TAP-CT block 12, SPECTRE block 24, VoCo stage 4 at stride 32), upsampled trilinearly. The decoder projects each of the four grids to 128 channels with a $1 \times 1 \times 1$ convolution, resizes them to a common grid (the token grid for the two ViTs, the stride-4 grid for VoCo), concatenates them (512 channels), fuses them with a 3^3 convolution, instance normalisation and leaky ReLU to 128 channels, then applies stages of trilinear $\times 2$ upsampling followed by the same convolution block that halves the channels (to a minimum of 16) until the voxel grid is reached (three stages for TAP-CT, four for SPECTRE, two for VoCo), and ends with a $1 \times 1 \times 1$ convolution to one logit per

structure. It has 2.45 M (TAP-CT), 2.62 M (SPECTRE) and 2.42 M (VoCo) parameters plus $17C$ or $33C$ for C structures, and no skip connections from the input.

Adapters. LoRA (Hu et al., 2022) replaces a frozen projection $\boldsymbol{W}_0\boldsymbol{x}$ by $\boldsymbol{W}_0\boldsymbol{x}+\frac{\alpha}{r}\boldsymbol{B}\boldsymbol{A}\,\mathrm{drop}(\boldsymbol{x})$ with $r = 16$, $\alpha = 32$ and dropout 0.1; $\boldsymbol{A}$ is initialised Kaiming-uniform and $\boldsymbol{B}$ at zero, so training starts from the pretrained model. It is applied to the fused query/key/value projection of every TAP-CT and VoCo block and to the separate query and key/value projections of every SPECTRE block, of SPECTRE's feature combiner, and of every Pillar-0 block. Late adapters cover TAP-CT blocks 9–12, SPECTRE blocks 17–24 and its feature combiner, the third stage of Pillar-0 (8 of 12 blocks) and the fourth stage of VoCo (2 of 8 blocks). rsLoRA scales the update by $\alpha/\sqrt{r}$ instead of α/r. DoRA (Liu et al., 2024) rescales each row of the adapted weight to a learned magnitude $\boldsymbol{m}$, $\boldsymbol{W} = \boldsymbol{m}\odot(\boldsymbol{W}_0+\frac{\alpha}{r}\boldsymbol{B}\boldsymbol{A})/\|\boldsymbol{W}_0+\frac{\alpha}{r}\boldsymbol{B}\boldsymbol{A}\|_{\mathrm{row}}$, with $\boldsymbol{m}$ initialised to the row norms of $\boldsymbol{W}_0$; the magnitude is a two-dimensional parameter and receives weight decay like the adapter matrices. Attention-only fine-tuning unfreezes every parameter of the attention modules that LoRA would adapt (query, key, value and output projections with their biases, and for VoCo the relative position bias). In segmentation the SPECTRE adapters and full fine-tuning also cover the feature combiner, which the segmentation path does not evaluate; those parameters receive no gradient.

Two-stage methods. Lin→LoRA and Attn→LoRA take the probe weights (for VoCo also the running standardisation statistics of the window tokens) from the parent run's checkpoint at step 4000 of its 8000-step schedule, add late-block LoRA with $\boldsymbol{B} = 0$, and train 4000 further steps with a fresh optimiser and schedule. Dec→LoRA takes the decoder from the Dec run at the midpoint of its schedule (step 4000; step 12 000 for the 24 000-step joint models) and trains all-block LoRA with the decoder for the second half.

Optimisation. AdamW uses $\beta = (0.9, 0.999)$ and $\epsilon = 10^{-8}$ (the framework defaults; no run overrides them). The learning rate rises linearly from 1% of its peak over the first 10% of the steps and then follows a cosine to 10^{-6}. Weight decay is applied to parameters of dimension two or more, excluding biases and normalisation weights. Gradients are not clipped. Classification trains at batch size 1 without gradient accumulation; segmentation at a micro-batch of 2 crops with accumulation 2, so four crops per step. The classification loss for a case with logits $\boldsymbol{z}$, labels $\boldsymbol{y}$ and known-label mask $\boldsymbol{m}$ is

$$\mathcal{L}_{\mathrm{cls}} = \frac{1}{B}\sum_{b=1}^{B}\sum_{t} m_{bt}\,\mathrm{BCE}(z_{bt}, y_{bt}; w_t), \qquad w_t = \mathrm{clip}\Big(\frac{n_t^-}{n_t^+}, 0.05, 20\Big), \tag{1}$$

where $\mathrm{BCE}(\cdot; w_t)$ is the binary cross-entropy with positive weight w_t computed from the negative and positive counts of label t in the training split. The segmentation loss per case b and structure c, with $\boldsymbol{p} = \sigma(\boldsymbol{z})$ over the V voxels of a crop and reference $\boldsymbol{y}$, is

$$\mathcal{L}_{\mathrm{seg}} = \frac{1}{BC}\sum_{b,c}\Big[1 - \frac{2\sum_v p_{bcv}y_{bcv} + 1}{\sum_v p_{bcv} + \sum_v y_{bcv} + 1} + \frac{1}{V}\sum_{v}\mathrm{BCE}(z_{bcv}, y_{bcv})\Big]. \tag{2}$$

In the joint CVPR26 models the three sources are sampled in the ratio 10:8:8 (AMOS, COVID-CTset, LUNA25); every other run cycles one shuffled training set.

Table 4: Classification settings per method. LoRA-type adapters have rank 16, $\alpha = 32$ and dropout 0.1. Learning rate and weight decay apply to every trained parameter unless a backbone/head pair is given. All runs: AdamW, 8000 steps (two-stage: 4000 + 4000), batch size 1, warm-up 10%, cosine to 10^{-6}, bf16, head dropout 0.2.

Method	Trained in the encoder	Head	Learning rate / weight decay
Lin	nothing	linear	10^{-3} / 0
Attn	nothing	attentive	10^{-4} / 0.01
LoRA	LoRA on q, k, v of every block	attentive	10^{-4} / 0.01
$\text{LoRA}_{\text{late}}$	LoRA on q, k, v of the late blocks	attentive	10^{-4} / 0.01
LoRA+ABMIL	as LoRA	ABMIL then attentive	10^{-4} / 0.01
DoRA	DoRA on q, k, v of every block	attentive	10^{-4} / 0.01
rsLoRA	LoRA, scale $\alpha/\sqrt{r}$	attentive	10^{-4} / 0.01
Lin→LoRA	late LoRA from step 4000 of Lin	linear (inherited)	10^{-4} / 0.01
Attn→LoRA	late LoRA from step 4000 of Attn	attentive (inherited)	10^{-4} / 0.01
Attn-only FT	all attention parameters	attentive	encoder 2×10^{-5} / 0.05, head 10^{-4} / 0.01
Full FT	every parameter	attentive	encoder 2×10^{-5} / 0.05, head 10^{-4} / 0.01
Repeats and controls (Appendix D)			
Full FT, lower LR	every parameter	attentive	encoder 5×10^{-6} / 0.05, head 10^{-4} / 0.01
Full FT, layer decay	every parameter, LR $\times 0.75$ per block from the top	attentive	encoder 2×10^{-5} / 0.05, head 10^{-4} / 0.01
Full FT, accumulation	every parameter; accumulation 8, clipping 1.0, 1000 steps	attentive	as Full FT
MLP-only FT	both MLP layers of every block	attentive	encoder 2×10^{-5} / 0.05, head 10^{-4} / 0.01
rsLoRA, matched LR	LoRA, scale $\alpha/\sqrt{r}$	attentive	2.5×10^{-5} / 0.01

Table 5: Segmentation settings per method. All runs: AdamW, 8000 steps (24 000 for the AMOS22 and TotalSegmentator joint models; two-stage methods split the budget in halves), four crops per step, warm-up 10%, cosine to 10^{-6}, bf16. LoRA-type adapters as in Table 4.

Method	Trained in the encoder	Head	Learning rate / weight decay
Lin	nothing	linear probe	10^{-2} / 0
Dec	nothing	decoder	10^{-3} / 10^{-4}
LoRA	LoRA on q, k, v of every block	decoder	10^{-4} / 10^{-4}
$\text{LoRA}_{\text{late}}$	LoRA on q, k, v of the late blocks	decoder	10^{-4} / 10^{-4}
DoRA	DoRA on q, k, v of every block	decoder	10^{-4} / 10^{-4}
Full FT	every parameter	decoder	encoder 2×10^{-5}, head 10^{-3} / 10^{-4}
Dec→LoRA	all-block LoRA from the midpoint of Dec	decoder (inherited)	10^{-4} / 10^{-4}
Controls (Appendix D)			
From scratch	every parameter, encoder re-initialised	decoder	as Full FT
Compute-matched per-structure	Dec, LoRA, Full FT at 1600 steps per structure	decoder	as above

Model selection and per-case outputs. Every 1000 steps the selection and held-out splits are evaluated and a checkpoint of the trained parameters is written; the checkpoint with the highest selection macro AUROC (mean over labels with both classes present) or mean Dice (mean over structures of the mean over cases in which the structure is present in the reference or the prediction) is kept, ties keeping the earlier checkpoint. For TotalSegmentator only the selection split is evaluated during training; the held-out split is evaluated afterwards with the selected checkpoint, and the recomputed selection Dice is required to match the recorded value within 0.005. Per case, the held-out probability of every known label (classification) and the Dice of every structure (segmentation) at the selected checkpoint are stored and are the input of every number in this paper. Dice thresholds the logits at 0; a structure enters a case's mean when the reference or the prediction contains it, so a false-positive structure scores 0 and an absent, unpredicted structure is skipped.

Appendix C. Statistical procedure

Every number in this paper is computed by one script from the stored per-case predictions of the selected checkpoint of every run (2388 runs; the classification files hold the held-out probability of every known label of every case, the segmentation files the Dice of every structure of every case).

Per-run metrics. AUROC is the Mann–Whitney statistic with midranks (tied scores count one half); a label is skipped for a run when its selection or held-out split has fewer than two cases or a single class, which did not occur. Dice per structure is the mean over the held-out cases in which the structure is present in the reference or the prediction; an absent, unpredicted structure is skipped and a false-positive structure counts as 0. Cohort and dataset means (Tables 16 and 17) are means over the labels or structures of complete cells. The recomputed selection-split macro metric was compared with the value logged during training for every run: all 1354 classification runs agree to 10^{-7}, and all but one of the 1034 segmentation runs within 2×10^{-3} (108 segmentation runs had their per-case Dice re-inferred from the selected checkpoint after training; the one exception differs by 2.01×10^{-3}).

Did-not-train rule. For each backbone and method, the median over label–mode pairs of the standard deviation of the held-out probabilities is computed; a median below 0.010 flags a method that did not train. Only SPECTRE Full FT is flagged (0.004; every other method on TAP-CT, SPECTRE and Pillar-0 is at least 0.114). It stays in the per-cohort tables, the paired tests and the Full FT − Lin estimate, and leaves the ranks and every average over methods. VoCo spreads are reported (Table 15) but the rule is not applied to VoCo, whose methods have small spreads at held-out AUROCs well above chance.

Paired tests. Per backbone, cohort, mode (per-task, joint) and label, every method is compared with the baselines Lin, LoRA, Attn and Full FT (classification; two-sided DeLong test of the AUROC difference on the same cases (DeLong et al., 1988)) or Lin, Dec and LoRA (segmentation; paired case bootstrap of the mean Dice difference, 4000 resamples shared by every comparison of the same group, two-sided bootstrap p). Pairs require identical case sets; none had to be skipped. p-values are Benjamini–Hochberg adjusted (Benjamini and Hochberg, 1995) within each family of comparisons (task, baseline, VoCo or not), and

"significant" means an adjusted p below 0.05; 7520 DeLong tests and 3591 bootstrap tests were made.

Label-level estimates. The unit is one label (structure) on one backbone. For a comparison A − B its value is the mean over the per-task and joint modes of the difference of the two methods' held-out metric (both modes must exist); for joint minus per-task it is the mean over the methods trained in both modes. The reported mean is over the labels of a backbone, or pooled over TAP-CT, SPECTRE and Pillar-0 (78 label–backbone units in classification, 74 in segmentation). Its 95% interval is a percentile bootstrap with 4000 resamples that draws cohort–backbone clusters with replacement, then labels within each drawn cluster, and then replaces each drawn label's value by one of 1000 case-level resamples of that label (positives and negatives resampled separately in classification, with the same case indices for every term of the difference; all cases in segmentation). Besides the comparisons with the linear probe (Lin) and the frozen decoder (Dec), every method with the attentive head is compared with the frozen attentive probe (matched head), and LoRA − Lin is also computed without FDG-PET/CT and the COVID-CTset label. Table 20 also gives the interval without the case stage. The number of cohort–backbone clusters with a positive mean is reported next to every estimate.

Ranks. A slot is a label (structure) on one backbone in one mode with every method of the task present (minus methods that did not train); methods are ranked by held-out AUROC or Dice with ties sharing the average rank. Classification has 156 slots over the three main backbones and 32 on VoCo; segmentation 118 and 59 (TotalSegmentator per-structure slots lack three methods and are not ranked).

Seeds, controls and convergence. Seed repeats are compared through cohort means per seed and the number of cohort–seed pairs in which one method is ahead of another; full fine-tuning variants through the selection-split AUROC, the number of cohorts whose run trained (median spread at least 0.010) and, for the variant with the highest selection AUROC, its label-level difference to Lin and LoRA; compute-matched models through the mean over structures of joint minus matched Dice. Convergence is the share of runs whose selected checkpoint is the last or one of the last two validations. All random draws use fixed seeds, so the script is deterministic.

Appendix D. Repeats and controls

Appendix E. VoCo

Appendix F. Ranks, paired tests and per-label results

F.1. Per-label and per-structure results

Held-out AUROC per label and Dice per structure for every method, per backbone, cohort or dataset and training mode. Best results are bolded, while second-best results are underlined.

Table 6: Seed repeats of the classification joint models: held-out AUROC (mean over the labels of a cohort) for seeds 42, 43 and 44 per backbone, cohort and method, with the standard deviation across seeds.

Backbone	Cohort	Lin				LoRA				Attn-only FT				Full FT			
		42	43	44	SD	42	43	44	SD	42	43	44	SD	42	43	44	SD
TAP-CT	CVPR26	0.707	0.700	0.708	0.004	0.733	0.723	0.701	0.016	0.739	0.732	0.750	0.009	0.744	0.699	0.691	0.029
	FDG-PET/CT	0.858	0.859	0.864	0.003	0.866	0.855	0.870	0.008	0.870	0.867	0.888	0.012	0.608	0.809	0.567	0.130
	RSNA 2022 C-spine	0.626	0.628	0.616	0.006	0.714	0.648	0.616	0.050	0.712	0.687	0.675	0.019	0.732	0.690	0.662	0.035
	RSNA 2023 trauma	0.739	0.746	0.731	0.008	0.757	0.762	0.760	0.002	0.833	0.875	0.817	0.030	0.881	0.612	0.888	0.157
	RSNA 2019 hemorrhage	0.871	0.870	0.874	0.002	0.906	0.889	0.894	0.009	0.915	0.882	0.914	0.019	0.854	0.742	0.910	0.086
SPECTRE	CVPR26	0.702	0.714	0.717	0.008	0.716	0.765	0.740	0.024	0.740	0.784	0.753	0.023	0.575†	–	–	–
	FDG-PET/CT	0.884	0.890	0.887	0.003	0.901	0.901	0.901	0.000	0.906	0.917	0.906	0.006	0.673†	–	–	–
	RSNA 2022 C-spine	0.629	0.606	0.541	0.046	0.706	0.687	0.694	0.009	0.718	0.707	0.623	0.052	0.554†	–	–	–
	RSNA 2023 trauma	0.789	0.797	0.790	0.004	0.846	0.847	0.844	0.001	0.803	0.839	0.860	0.028	0.654†	–	–	–
Pillar-0	RSNA 2019 hemorrhage	0.888	0.888	0.888	0.000	0.912	0.917	0.920	0.004	0.912	0.923	0.927	0.007	0.878	0.896	0.907	0.015

Table 7: Seed repeats of the segmentation joint models: held-out Dice (mean over structures) for seeds 42, 43 and 44 per backbone, dataset and method, with the standard deviation across seeds.

Backbone	Dataset	Dec				LoRA				Full FT			
		42	43	44	SD	42	43	44	SD	42	43	44	SD
TAP-CT	AMOS22	0.708	0.661	0.655	0.029	0.739	0.747	0.642	0.059	0.681	0.681	0.679	0.001
	KiTS23	0.514	0.519	0.515	0.003	0.582	0.573	0.572	0.005	0.519	0.524	0.545	0.014
	LiTS	0.699	0.716	0.730	0.016	0.715	0.717	0.724	0.004	0.715	0.724	0.711	0.007
	MSD Pancreas	0.480	0.478	0.483	0.002	0.505	0.495	0.487	0.009	0.487	0.480	0.437	0.027
	TotalSegmentator	0.798	0.752	0.798	0.026	0.776	0.765	0.780	0.008	0.815	0.814	0.815	0.000
SPECTRE	AMOS22	0.658	0.658	0.659	0.000	0.784	0.762	0.779	0.011	0.669	0.666	0.672	0.003
	KiTS23	0.572	0.591	0.576	0.010	0.588	0.628	0.631	0.024	0.519	0.546	0.521	0.015
	LiTS	0.742	0.741	0.740	0.001	0.753	0.729	0.736	0.012	0.713	0.722	0.685	0.019
	MSD Pancreas	0.514	0.519	0.521	0.004	0.550	0.559	0.544	0.008	0.530	0.513	0.508	0.011
	TotalSegmentator	0.770	0.726	0.728	0.025	0.733	0.750	0.773	0.020	0.795	0.747	0.745	0.028

Table 8: Full fine-tuning settings and controls on the classification joint models: held-out AUROC per backbone and cohort for full fine-tuning with the default setting, a lower learning rate (5×10^{-6}), layer-wise learning-rate decay (0.75 per block) and gradient accumulation with clipping, for MLP-only fine-tuning and for rsLoRA at a quarter of the learning rate, next to Lin and LoRA.

Backbone	Cohort (labels)	Lin	LoRA	Full FT	Full FT, lower LR	Full FT, layer decay	Full FT, accumulation	MLP-only FT	rsLoRA	rsLoRA, matched LR
TAP-CT	CVPR26 (17)	0.707	0.733	0.744	0.730	**0.750**	0.685	0.745	0.722	0.741
	FDG-PET/CT (4)	0.858	0.866	0.608	0.861	**0.867**	**0.867**	0.866	0.785	0.855
	RSNA 2022 C-spine (8)	0.626	0.714	**0.732**	0.654	0.618	0.687	0.627	0.601	0.611
	RSNA 2023 trauma (4)	0.739	0.757	0.881	0.860	0.869	0.758	**0.894**	0.684	0.751
	RSNA 2019 hemorrhage (6)	0.871	0.906	0.854	0.889	**0.917**	0.909	0.907	0.869	0.901
SPECTRE	CVPR26 (17)	0.702	0.716	0.575†	0.713	**0.758**	0.511	0.681	0.690	0.739
	FDG-PET/CT (4)	0.884	0.901	0.673†	0.884	0.865	0.883	0.814	0.900	**0.908**
	RSNA 2022 C-spine (8)	0.629	**0.706**	0.554†	0.568	0.700	0.553	0.626	0.691	0.672
	RSNA 2023 trauma (4)	0.789	0.846	0.654†	0.793	**0.863**	0.578	0.838	0.850	0.834
Pillar-0	RSNA 2019 hemorrhage (6)	0.888	0.912	0.878	**0.924**	0.920	0.918	0.912	0.918	0.920

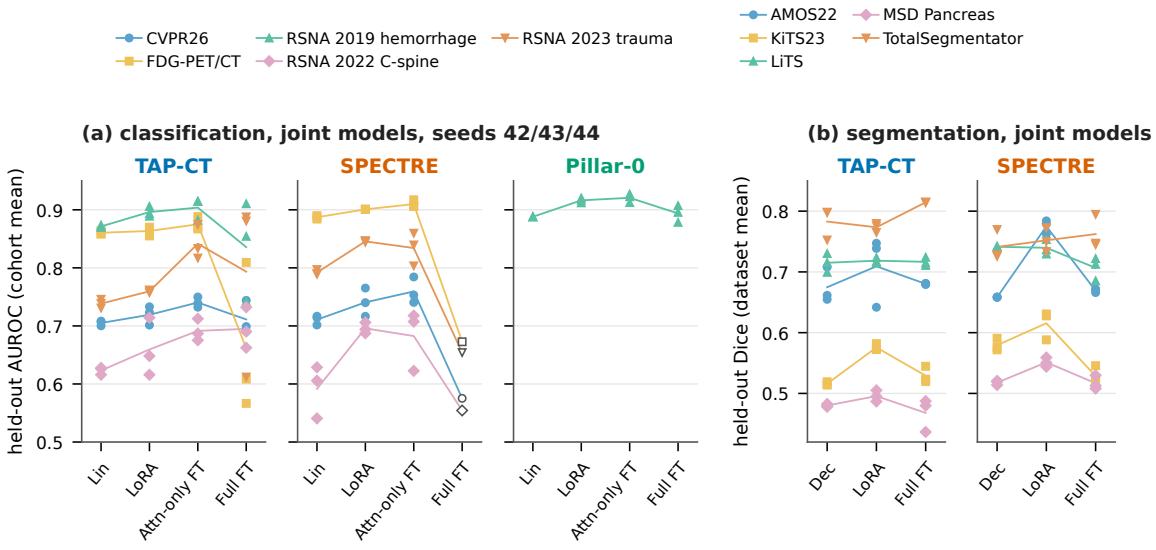


Figure 6: Between-seed variation of the joint models next to between-method differences: cohort (dataset) means of the three seeds per method, (a) classification, (b) segmentation.

Table 9: Compute-matched segmentation: dataset-mean held-out Dice of the 24 000-step joint model, of per-structure models trained for 1600 steps each (the same total number of steps) and their difference, for Dec, LoRA and Full FT on AMOS22 and TotalSegmentator.

Dataset (structures)	Backbone	Method	Joint 24 000 steps	Per-task, matched 1600 steps each	Joint − matched	Per-task 8000 steps each
AMOS22 (15)	TAP-CT	Dec	0.708	0.763	-0.055	0.793
		LoRA	0.739	0.617	+0.122	0.767
		Full FT	0.681	0.766	-0.085	0.805
	SPECTRE	Dec	0.658	0.760	-0.101	0.808
		LoRA	0.784	0.648	+0.136	0.812
		Full FT	0.669	0.793	-0.123	0.792
TotalSegmentator (15)	TAP-CT	Dec	0.798	0.783	+0.015	0.843
		LoRA	0.776	0.657	+0.119	0.819
		Full FT	0.815	0.803	+0.012	0.832
	SPECTRE	Dec	0.770	0.806	-0.037	0.848
		LoRA	0.733	0.691	+0.042	0.848
		Full FT	0.795	0.819	-0.024	0.838

Table 10: From-scratch control: TAP-CT architecture with a randomly initialised encoder, trained as the Full FT joint model, next to Full FT and Dec, per dataset (held-out Dice, mean over structures).

Dataset (structures)	From scratch	Dec	Full FT	Scratch − Dec	Scratch − Full FT
AMOS22 (15)	0.496	0.708	0.681	-0.212	-0.185
KiTS23 (3)	0.400	0.514	0.519	-0.114	-0.120
LiTS (2)	0.612	0.699	0.715	-0.087	-0.103
MSD Pancreas (2)	0.308	0.480	0.487	-0.172	-0.180
TotalSegmentator (15)	0.579	0.798	0.815	-0.219	-0.236

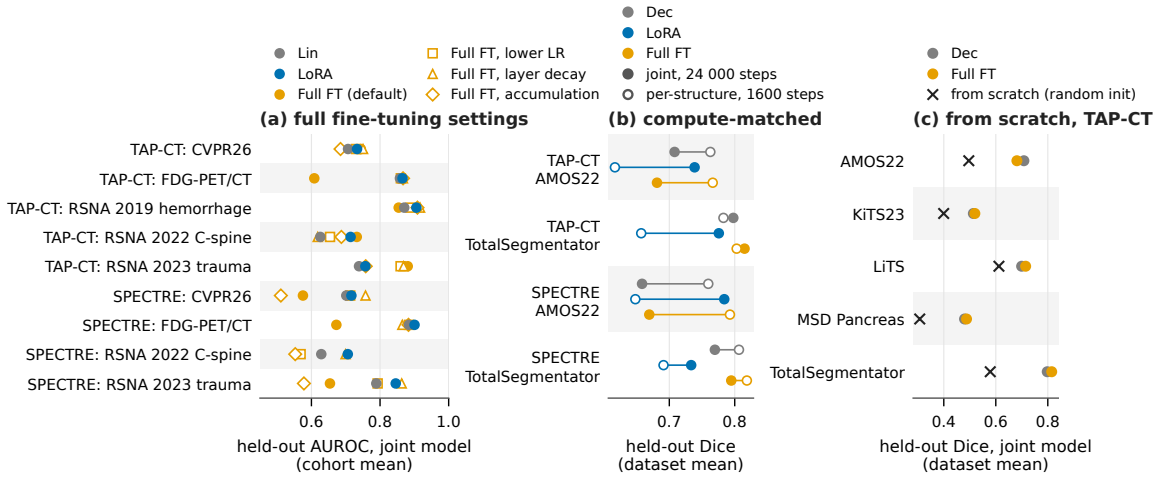


Figure 7: Repeats and controls. (a) Full fine-tuning settings against Lin and LoRA on the classification joint models. (b) Compute-matched segmentation: joint model against per-structure models at 1600 steps. (c) From-scratch control against Full FT and Dec on TAP-CT.

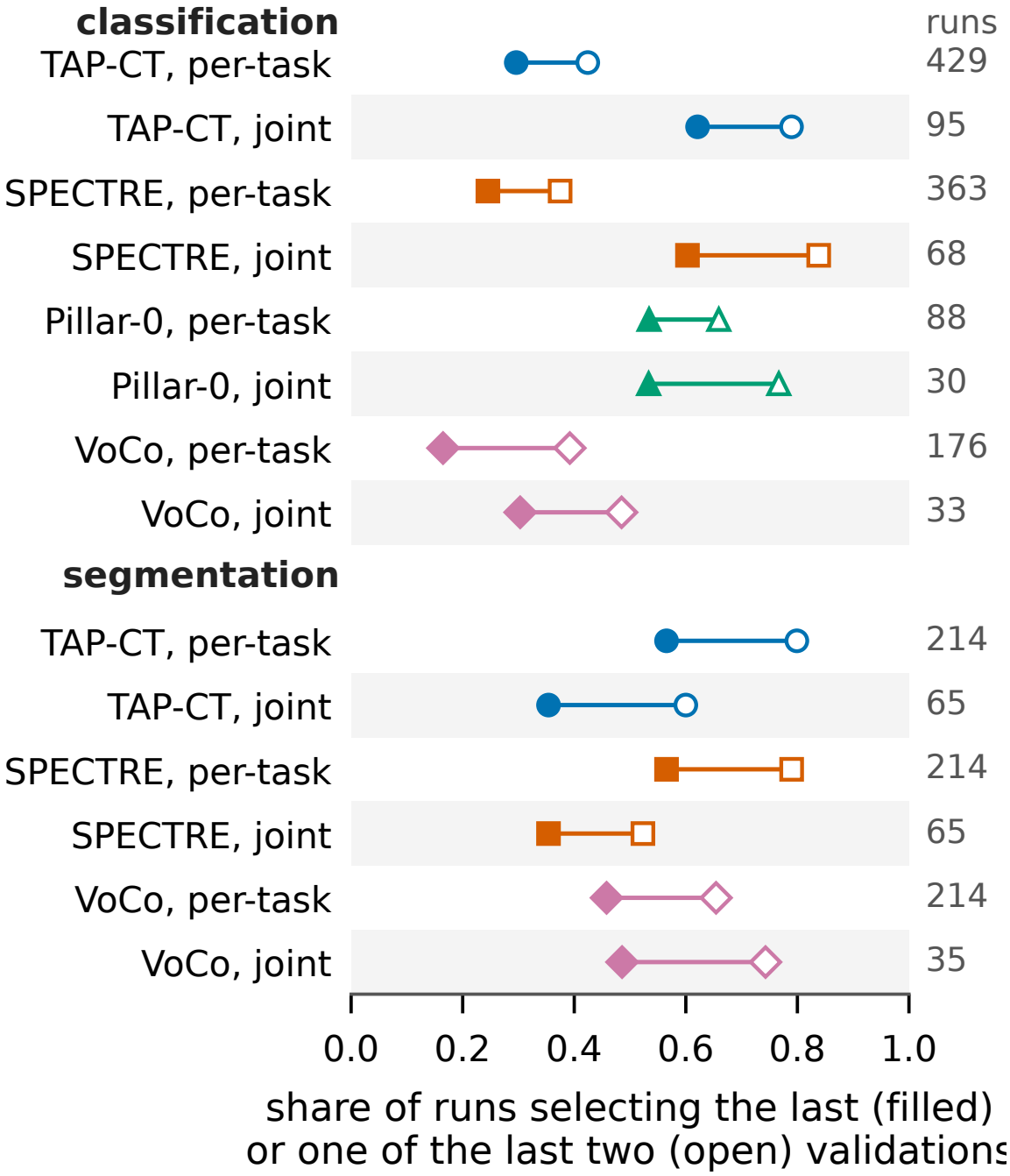


Figure 8: Share of runs whose selected checkpoint is the last validation or one of the last two, per task and backbone.

Table 11: Convergence: share of runs that selected the last validation or one of the last two, per task and backbone.

Backbone	Mode	Runs	Last validation selected (%)	One of last two selected (%)
Classification				
TAP-CT	per-task	429	30	42
	joint	120	64	79
	joint (26 000 steps)	11	9	9
SPECTRE	per-task	363	25	37
	joint	88	56	78
	joint (26 000 steps)	11	18	27
Pillar-0	per-task	88	53	66
	joint	35	60	80
VoCo	per-task	176	16	39
	joint	33	30	48
All	all	1354	33	49
Segmentation				
TAP-CT	per-task	214	57	80
	joint	70	34	63
	joint (8000 steps)	14	79	93
	per-task, compute-matched	90	49	74
SPECTRE	per-task	214	57	79
	joint	65	35	52
	joint (8000 steps)	14	93	100
	per-task, compute-matched	90	66	90
VoCo	per-task	214	46	65
	joint	35	49	74
	joint (8000 steps)	14	64	86
All	all	1034	52	75

Table 12: The comparisons of Table 1 on VoCo (16 labels from FDG-PET/CT, RSNA 2022 and RSNA 2023; all 37 structures).

Comparison	VoCo	Cohorts > 0
Classification (AUROC)		
LoRA − Lin	-0.071 [-0.168, -0.005]	0/3
LoRA − Attn	-0.031 [-0.145, +0.043]	1/3
Attn-only FT − LoRA	+0.036 [-0.024, +0.102]	2/3
Full FT − Lin	-0.034 [-0.136, +0.027]	1/3
joint − per-task	-0.001 [-0.022, +0.018]	2/3
Segmentation (Dice)		
Dec − Lin	+0.680 [+0.433, +0.746]	5/5
LoRA − Dec	-0.018 [-0.047, +0.016]	2/5
LoRA − Dec, per-task	-0.035 [-0.062, -0.007]	0/5
LoRA − Dec, joint	0.000 [-0.034, +0.047]	2/5
Full FT − Dec	+0.015 [+0.007, +0.023]	4/5
joint − per-task	-0.041 [-0.081, +0.004]	2/5

Table 13: VoCo classification: held-out AUROC as in Table 16.

Cohort (labels)	Backbone	Mode	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ ABMIL	DoRA	rsLoRA	Lin→ LoRA	Attn→ LoRA	Attn-only FT	Full FT
FDG-PET/CT (4)	VoCo	per-task	0.720	0.685	0.512	0.671	0.562	0.669	0.598	**0.736**	0.591	0.665	0.606
		joint	0.745	0.739	0.600	0.506	0.655	0.564	0.527	**0.750**	0.520	0.641	0.582
RSNA 2022 C-spine (8)	VoCo	per-task	0.606	0.508	0.537	0.551	0.596	0.561	**0.617**	0.602	0.521	0.576	0.605
		joint	0.586	0.548	0.560	0.550	0.565	0.589	0.578	**0.619**	0.527	0.589	0.593
RSNA 2023 trauma (4)	VoCo	per-task	0.617	0.625	0.622	0.601	0.595	0.611	0.608	**0.631**	0.569	0.588	0.596
		joint	0.631	0.616	0.603	0.627	0.570	0.583	0.607	0.626	0.635	0.593	**0.648**

Table 14: VoCo segmentation: held-out Dice as in Table 17.

Dataset (structures)	Backbone	Mode	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→ LoRA
AMOS22 (15)	VoCo	per-task	0.041	0.805	0.782	0.783	0.783	**0.818**	0.799
		joint	0.038	0.739	0.762	0.758	**0.764**	0.752	0.736
KiTS23 (3)	VoCo	per-task	0.017	0.519	0.501	0.500	0.493	**0.564**	0.481
		joint	0.037	0.465	0.501	**0.516**	0.507	0.481	0.452
LiTS (2)	VoCo	per-task	0.342	0.733	0.692	0.694	0.688	**0.748**	0.720
		joint	0.345	0.732	0.720	0.710	0.718	**0.737**	0.726
MSD Pancreas (2)	VoCo	per-task	0.000	**0.569**	0.539	0.545	0.541	0.566	0.558
		joint	0.000	0.559	0.552	0.545	0.552	0.560	**0.561**
TotalSegmentator (15)	VoCo	per-task	0.055	0.822	0.771	–	–	**0.838**	–
		joint	0.047	0.732	0.705	0.697	0.709	**0.750**	0.723

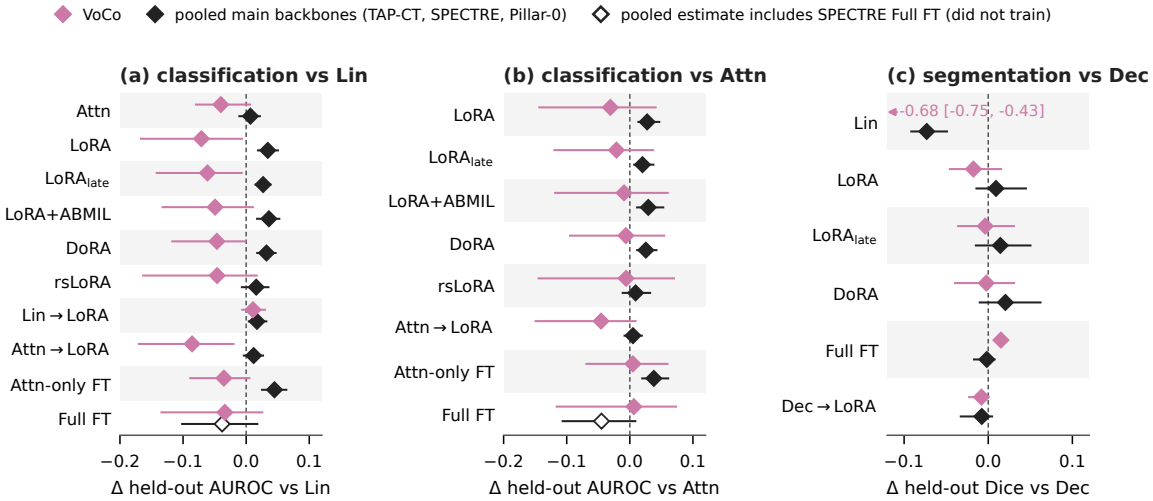


Figure 9: VoCo. (a) Label-level difference of every classification method to the linear probe on VoCo, next to the estimate pooled over the main backbones. (b) Per-structure change in held-out Dice relative to the frozen decoder on VoCo.

Table 15: Median over label–mode pairs of the standard deviation of the held-out probabilities per backbone and method (the did-not-train rule flags values below 0.010; applied to TAP-CT, SPECTRE and Pillar-0 only).

Backbone	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ABMIL	DoRA	rsLoRA	Lin→LoRA	Attn→LoRA	Attn-only FT	Full FT
TAP-CT	0.227	0.239	0.239	0.235	0.233	0.237	0.167	0.222	0.225	0.247	0.170
SPECTRE	0.187	0.114	0.161	0.185	0.187	0.185	0.150	0.178	0.129	0.148	**0.004**†
Pillar-0	0.294	0.321	0.359	0.350	0.352	0.354	0.359	0.337	0.338	0.355	0.347
VoCo	0.152	0.023	0.035	0.032	0.008	0.030	0.036	0.150	0.020	0.041	0.036

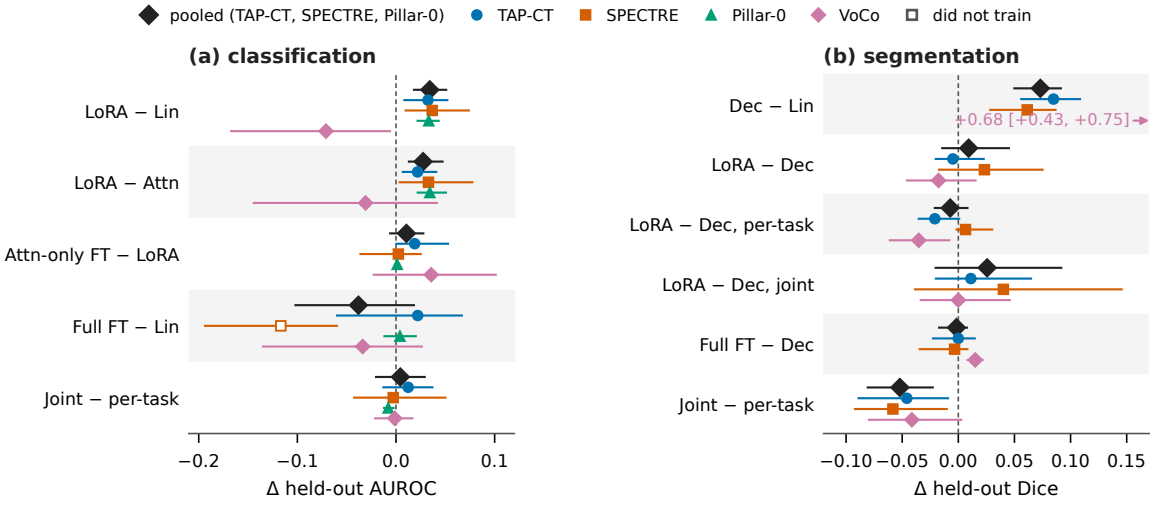


Figure 10: The comparisons of Table 1 per backbone: label-level mean difference in held-out AUROC (a) or Dice (b) with its 95% interval; 0 means no difference. The black diamond pools TAP-CT, SPECTRE and Pillar-0; VoCo has its own marker and is never pooled.

Table 16: Classification: held-out AUROC, mean over the labels of each cohort (number of labels in brackets), per backbone, training mode and method. Best results are bolded, while second-best results are underlined. A shaded cell marks the better of per-task and joint training for that method, backbone and cohort. SPECTRE Full FT did not train with the default setting (Section 4.1) and is not marked.

Cohort (labels)	Backbone	Mode	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ ABMIL	DoRA	rsLoRA	Lin→ LoRA	Attn→ LoRA	Attn-only FT	Full FT
CVPR26 (17)	TAP-CT	per-task	0.677	0.718	0.732	0.704	0.728	0.717	0.724	0.697	0.707	**0.744**	0.720
		joint	0.707	0.720	0.733	0.726	0.729	0.733	0.722	0.695	0.708	0.739	**0.744**
	SPECTRE	per-task	0.736	0.725	0.769	0.769	0.776	0.766	0.768	0.769	0.765	**0.778**	0.627†
		joint	0.702	0.738	0.716	0.731	**0.747**	0.731	0.690	0.744	0.728	0.740	0.575†
FDG-PET/CT (4)	TAP-CT	per-task	0.863	0.836	0.852	**0.868**	0.839	0.847	0.817	0.866	0.856	0.832	0.769
		joint	0.858	0.860	0.866	0.864	0.849	0.858	0.785	0.860	0.862	**0.870**	0.608
	SPECTRE	per-task	0.857	0.862	0.882	0.879	0.874	**0.892**	0.872	0.887	0.867	0.848	0.617†
		joint	0.884	0.880	0.901	0.888	0.889	0.900	0.900	0.884	0.884	**0.906**	0.673†
RSNA 2022 C-spine (8)	TAP-CT	per-task	0.619	0.625	0.623	0.608	0.613	0.641	0.635	0.606	0.629	**0.660**	0.637
		joint	0.626	0.624	0.714	0.706	0.713	0.680	0.601	0.624	0.620	0.712	**0.732**
	SPECTRE	per-task	0.579	0.591	0.635	**0.654**	0.644	0.619	0.638	0.615	0.591	0.635	0.566†
		joint	0.629	0.601	0.706	0.644	**0.730**	0.692	0.691	0.670	0.611	0.718	0.554†
RSNA 2023 trauma (4)	TAP-CT	per-task	0.737	0.718	0.739	0.756	0.703	0.752	0.725	0.738	0.716	**0.859**	0.775
		joint	0.739	0.738	0.757	0.744	0.758	0.768	0.684	0.746	0.758	0.833	**0.881**
	SPECTRE	per-task	0.791	0.778	0.824	0.832	**0.852**	0.842	0.836	0.825	0.798	0.772	0.624†
		joint	0.789	0.788	0.846	0.835	**0.859**	0.838	0.850	0.827	0.810	0.803	0.654†
RSNA 2019 hemorrhage (6)	TAP-CT	per-task	0.870	0.873	0.894	0.894	0.891	0.897	0.870	0.877	0.878	**0.911**	0.888
		joint	0.871	0.877	0.906	0.895	0.886	0.897	0.869	0.868	0.876	**0.915**	0.854
	Pillar-0	per-task	0.886	0.884	0.929	0.914	0.902	**0.931**	0.920	0.912	0.903	**0.931**	0.904
		joint	0.888	0.888	0.912	0.892	**0.925**	0.912	0.918	0.905	0.898	0.912	0.878

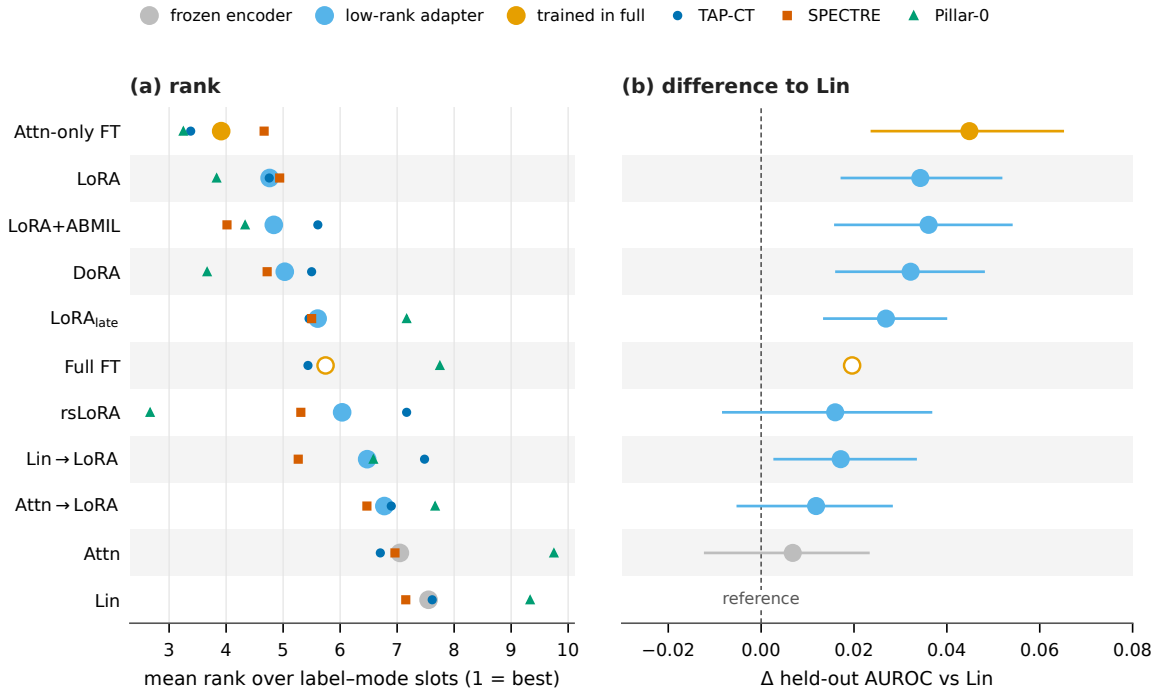


Figure 11: Classification methods on the three main backbones. (a) Mean rank over label–mode pairs, 1 = best; large markers pool the backbones, small markers are per backbone. (b) Label-level mean difference to the linear probe with its 95% interval. SPECTRE Full FT, which did not train, is excluded from the ranks and drawn open in (b).

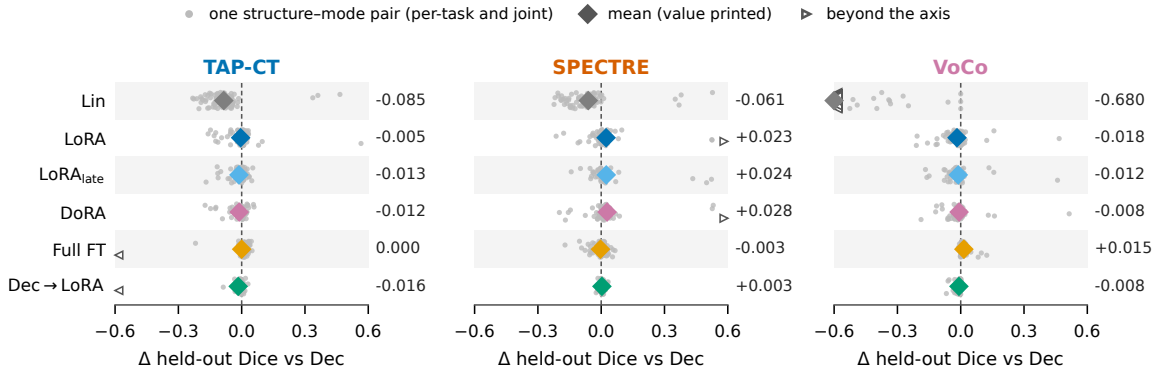


Figure 12: Per-structure change in held-out Dice relative to the frozen decoder of the same backbone, dataset and mode (grey: structure–mode pairs; coloured: mean, printed on the right; open triangles: beyond the axis), for TAP-CT, SPECTRE and VoCo.

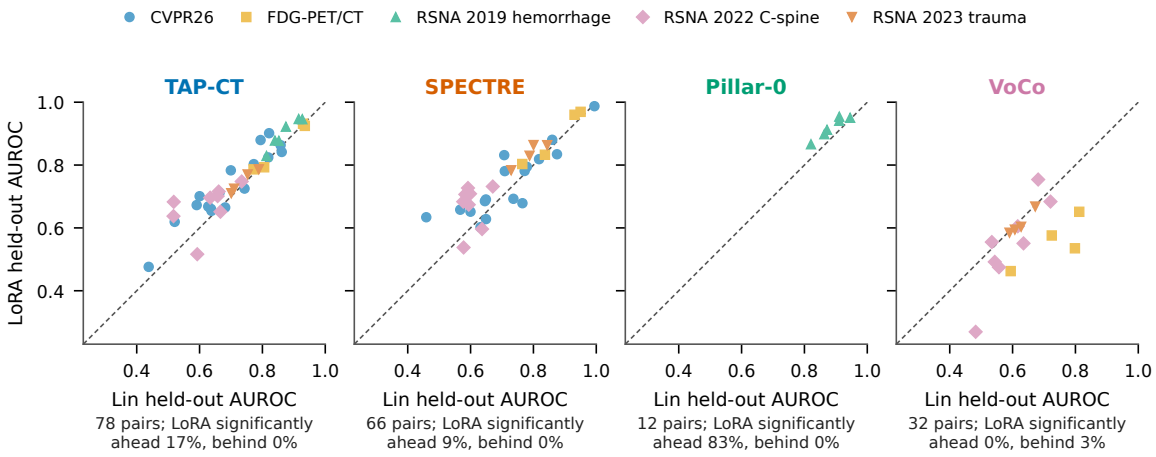


Figure 13: Held-out AUROC of LoRA against the linear probe per label and mode, one panel per backbone; points above the identity line favour LoRA. The text under each panel gives the share of label–mode pairs in which LoRA is significantly ahead or behind.

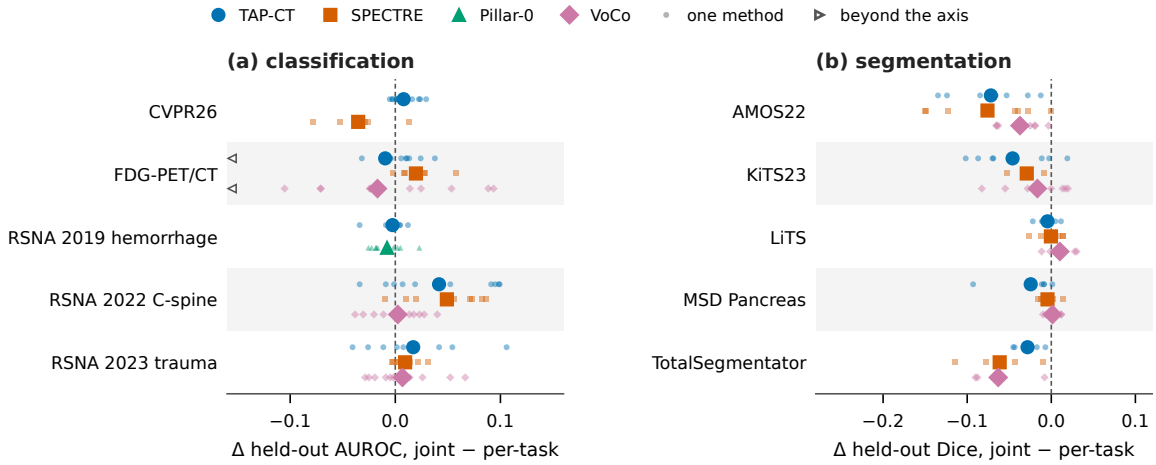


Figure 14: Joint minus per-task held-out AUROC per classification cohort (a) and joint minus per-structure held-out Dice per segmentation dataset (b), per backbone; small markers are methods, large markers their mean.

Table 17: Segmentation: held-out Dice, mean over the structures of each dataset (number of structures in brackets), per backbone, training mode and method; – marks methods not run per structure on TotalSegmentator. Best results are bolded, while second-best results are underlined. A shaded cell marks the better of per-structure and joint training for that method, backbone and dataset. The AMOS22 and TotalSegmentator joint models train 24 000 steps; their 8000-step runs are in Table 23.

Dataset (structures)	Backbone	Mode	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→ LoRA
AMOS22 (15)	TAP-CT	per-task	0.658	0.793	0.767	0.764	0.761	**0.805**	0.791
		joint	0.646	0.708	**0.739**	0.698	0.708	0.681	0.656
	SPECTRE	per-task	0.683	0.808	0.812	0.806	**0.814**	0.792	**0.814**
		joint	0.682	0.658	**0.784**	0.764	0.775	0.669	0.664
KiTS23 (3)	TAP-CT	per-task	0.531	0.584	0.563	0.567	0.563	**0.588**	0.587
		joint	0.430	0.514	**0.582**	0.556	0.560	0.519	0.500
	SPECTRE	per-task	0.583	0.599	0.613	0.617	**0.622**	0.528	0.603
		joint	0.530	0.572	0.588	0.588	**0.593**	0.519	0.570
LiTS (2)	TAP-CT	per-task	0.659	0.709	**0.716**	0.709	0.714	0.710	0.714
		joint	0.656	0.699	0.715	**0.721**	0.692	0.715	0.703
	SPECTRE	per-task	0.688	0.741	**0.756**	0.723	**0.756**	0.739	0.729
		joint	0.696	0.742	**0.753**	0.736	0.743	0.713	0.743
MSD Pancreas (2)	TAP-CT	per-task	0.427	0.491	0.504	0.501	0.502	**0.512**	0.495
		joint	0.334	0.480	**0.505**	0.492	0.493	0.487	0.468
	SPECTRE	per-task	0.475	0.516	**0.563**	0.538	0.557	0.516	0.518
		joint	0.459	0.514	**0.550**	0.529	**0.550**	0.530	0.520
TotalSegmentator (15)	TAP-CT	per-task	0.747	**0.843**	0.819	–	–	0.832	–
		joint	0.740	0.798	0.776	0.776	0.776	**0.815**	0.790
	SPECTRE	per-task	0.731	**0.848**	**0.848**	–	–	0.838	–
		joint	0.722	0.770	0.733	0.752	0.736	**0.795**	0.772

Table 18: Classification: mean rank per method (1 = best) over the label–mode pairs with every method present, per backbone and pooled over the three main backbones; share of pairs within 0.01 AUROC of the best method and mean deficit to the best method.

Method	TAP-CT 78 slots	SPECTRE 66 slots	Pillar-0 12 slots	All 156 slots	Within 0.01 of best (%)	Regret all
Lin	7.62	7.15	9.33	7.55	7	0.081
Attn	6.71	6.96	9.75	7.05	10	0.074
LoRA	4.76	4.94	3.83	4.76	23	0.046
$\text{LoRA}_{\text{late}}$	5.46	5.50	7.17	5.61	17	0.054
LoRA+ABMIL	5.61	**4.02**	4.33	4.84	26	0.045
DoRA	5.50	4.72	3.67	5.03	19	0.049
rsLoRA	7.17	5.31	**2.67**	6.04	18	0.065
Lin→LoRA	7.48	5.27	6.58	6.47	11	0.064
Attn→LoRA	6.90	6.47	7.67	6.78	13	0.069
Attn-only FT	**3.38**	4.67	3.25	**3.91**	**38**	**0.036**
Full FT	5.44	–†	7.75	5.74†	31	0.055

Table 19: Segmentation: mean rank per method over the structure–mode pairs with all seven methods, per backbone and pooled.

Method	TAP-CT 59 slots	SPECTRE 59 slots	All 118 slots	Within 0.01 of best (%)	Regret all
Lin	6.49	6.25	6.37	4	0.116
Dec	3.70	4.35	4.03	21	0.051
LoRA	3.38	3.06	3.22	47	**0.037**
$\text{LoRA}_{\text{late}}$	4.03	3.92	3.97	31	0.046
DoRA	3.91	**2.86**	3.38	44	0.043
Full FT	**2.05**	3.84	**2.94**	**64**	0.051
Dec→LoRA	4.44	3.72	4.08	24	0.058

Table 20: Label-level estimates for every comparison and backbone: number of labels, mean difference, 95% interval over cohorts and labels, 95% interval over cohorts, labels and held-out cases, and cohort–backbone clusters with a positive mean.

Comparison	Backbone	n	Mean	95% CI (two-stage)	95% CI (three-stage)	Cohorts > 0
LoRA − Lin	TAP-CT	39	+0.032	[+0.009, +0.049]	[+0.007, +0.053]	4/5
	SPECTRE	33	+0.037	[+0.013, +0.068]	[+0.009, +0.075]	4/4
	Pillar-0	6	+0.033	[+0.021, +0.042]	[+0.021, +0.044]	1/1
	VoCo	16	-0.071	[-0.162, -0.011]	[-0.168, -0.005]	0/3
	All	78	+0.034	[+0.019, +0.049]	[+0.017, +0.052]	9/10
LoRA − Attn	TAP-CT	39	+0.022	[+0.009, +0.039]	[+0.006, +0.042]	5/5
	SPECTRE	33	+0.033	[+0.007, +0.074]	[+0.003, +0.078]	4/4
	Pillar-0	6	+0.034	[+0.021, +0.051]	[+0.021, +0.052]	1/1
	VoCo	16	-0.031	[-0.150, +0.033]	[-0.145, +0.043]	1/3
	All	78	+0.027	[+0.013, +0.047]	[+0.012, +0.048]	10/10
Attn-only FT − LoRA	TAP-CT	39	+0.019	[+0.002, +0.053]	[-0.001, +0.054]	4/5
	SPECTRE	33	+0.002	[-0.037, +0.023]	[-0.037, +0.026]	2/4
	Pillar-0	6	+0.001	[0.000, +0.003]	[-0.002, +0.004]	1/1
	VoCo	16	+0.036	[-0.020, +0.097]	[-0.024, +0.102]	2/3
	All	78	+0.011	[-0.005, +0.026]	[-0.007, +0.029]	7/10
Full FT − Lin	TAP-CT	39	+0.022	[-0.057, +0.064]	[-0.061, +0.068]	4/5
	SPECTRE	33	-0.117	[-0.195, -0.063]	[-0.195, -0.059]	0/4
	Pillar-0	6	+0.004	[-0.011, +0.019]	[-0.013, +0.021]	1/1
	VoCo	16	-0.034	[-0.133, +0.018]	[-0.136, +0.027]	1/3
	All	78	-0.038	[-0.101, +0.017]	[-0.103, +0.019]	5/10
joint − per-task	TAP-CT	39	+0.012	[-0.005, +0.032]	[-0.014, +0.038]	3/5
	SPECTRE	33	-0.003	[-0.035, +0.049]	[-0.044, +0.051]	3/4
	Pillar-0	6	-0.008	[-0.012, -0.003]	[-0.013, -0.001]	0/1
	VoCo	16	-0.001	[-0.017, +0.012]	[-0.022, +0.018]	2/3
	All	78	+0.004	[-0.017, +0.027]	[-0.021, +0.030]	6/10
Attn − Lin	TAP-CT	39	+0.011	[-0.013, +0.027]	[-0.014, +0.029]	3/5
	SPECTRE	33	+0.004	[-0.022, +0.030]	[-0.030, +0.035]	2/4
	Pillar-0	6	-0.001	[-0.012, +0.007]	[-0.014, +0.008]	0/1
	VoCo	16	-0.040	[-0.072, 0.000]	[-0.081, +0.008]	0/3
	All	78	+0.007	[-0.010, +0.021]	[-0.012, +0.023]	5/10
$\text{LoRA}_{\text{late}}$ − Lin	TAP-CT	39	+0.023	[+0.009, +0.036]	[+0.005, +0.039]	5/5
	SPECTRE	33	+0.034	[+0.016, +0.051]	[+0.011, +0.058]	4/4
	Pillar-0	6	+0.016	[+0.008, +0.024]	[+0.007, +0.025]	1/1
	VoCo	16	-0.061	[-0.130, -0.010]	[-0.143, -0.005]	0/3
	All	78	+0.027	[+0.016, +0.038]	[+0.013, +0.040]	10/10
LoRA+ABMIL − Lin	TAP-CT	39	+0.024	[-0.006, +0.042]	[-0.005, +0.045]	3/5
	SPECTRE	33	+0.052	[+0.025, +0.080]	[+0.022, +0.086]	4/4
	Pillar-0	6	+0.026	[+0.016, +0.037]	[+0.015, +0.038]	1/1
	VoCo	16	-0.049	[-0.134, -0.001]	[-0.134, +0.012]	0/3
	All	78	+0.036	[+0.018, +0.052]	[+0.016, +0.054]	8/10
DoRA − Lin	TAP-CT	39	+0.028	[+0.009, +0.042]	[+0.005, +0.046]	4/5
	SPECTRE	33	+0.037	[+0.014, +0.059]	[+0.009, +0.067]	4/4
	Pillar-0	6	+0.034	[+0.021, +0.045]	[+0.020, +0.047]	1/1
	VoCo	16	-0.046	[-0.112, -0.011]	[-0.119, 0.000]	0/3
	All	78	+0.032	[+0.019, +0.045]	[+0.016, +0.048]	9/10
rsLoRA − Lin	TAP-CT	39	+0.003	[-0.043, +0.029]	[-0.043, +0.031]	1/5
	SPECTRE	33	+0.028	[+0.004, +0.063]	[0.000, +0.070]	4/4
	Pillar-0	6	+0.032	[+0.021, +0.041]	[+0.020, +0.043]	1/1
	VoCo	16	-0.046	[-0.163, +0.006]	[-0.165, +0.018]	1/3
	All	78	+0.016	[-0.007, +0.035]	[-0.008, +0.037]	6/10
Lin→LoRA − Lin	TAP-CT	39	+0.001	[-0.006, +0.007]	[-0.008, +0.010]	4/5
	SPECTRE	33	+0.035	[+0.018, +0.050]	[+0.012, +0.055]	4/4
	Pillar-0	6	+0.022	[+0.013, +0.029]	[+0.013, +0.030]	1/1
	VoCo	16	+0.011	[-0.002, +0.025]	[-0.008, +0.031]	3/3
	All	78	+0.017	[+0.004, +0.032]	[+0.003, +0.034]	9/10
Attn→LoRA − Lin	TAP-CT	39	+0.008	[-0.009, +0.019]	[-0.011, +0.024]	3/5
	SPECTRE	33	+0.016	[-0.014, +0.041]	[-0.017, +0.046]	3/4
	Pillar-0	6	+0.013	[+0.007, +0.020]	[+0.006, +0.021]	1/1
	VoCo	16	-0.086	[-0.171, -0.022]	[-0.172, -0.018]	0/3
	All	78	+0.012	[-0.003, +0.026]	[-0.005, +0.028]	7/10

Table 21: Classification paired tests: share of label–mode pairs in which the method is ahead of the baseline, and significantly ahead or behind (DeLong, Benjamini–Hochberg, adjusted $p < 0.05$), per backbone.

Backbone	Method	n	vs Lin				vs LoRA				vs Attn				vs Full FT			
			Δ	>0	+	−	Δ	>0	+	−	Δ	>0	+	−	Δ	>0	+	−
TAP-CT	Lin	78	–	–	–	–	-0.032	22	0	17	-0.011	41	0	4	-0.022	28	12	17
	Attn	78	+0.011	59	4	0	-0.022	36	0	15	–	–	–	–	-0.011	40	10	18
	LoRA	78	+0.032	78	17	0	–	–	–	–	+0.022	64	15	0	+0.010	55	18	6
	$\text{LoRA}_{\text{late}}$	78	+0.023	69	14	1	-0.010	35	1	9	+0.012	62	10	1	+0.001	44	18	6
	LoRA+ABMIL	78	+0.024	68	12	1	-0.008	47	4	13	+0.014	60	12	1	+0.002	45	17	12
	DoRA	78	+0.028	68	17	0	-0.005	40	4	10	+0.017	65	17	0	+0.006	51	17	9
	rsLoRA	78	+0.003	55	5	5	-0.029	28	1	23	-0.008	44	5	6	-0.019	47	13	22
	Lin→LoRA	78	+0.001	51	5	1	-0.031	28	0	17	-0.009	44	3	4	-0.021	31	9	18
	Attn→LoRA	78	+0.008	56	9	0	-0.024	28	0	17	-0.003	47	3	0	-0.014	38	13	15
	Attn-only FT	78	+0.051	83	24	0	+0.019	64	17	0	+0.041	81	26	0	+0.029	63	23	5
	Full FT	78	+0.022	72	13	8	-0.010	45	6	17	+0.011	59	15	8	–	–	–	–
SPECTRE	Lin	66	–	–	–	–	-0.037	26	0	9	-0.004	50	2	3	–	–	–	–
	Attn	66	+0.004	50	3	2	-0.033	29	0	9	–	–	–	–	–	–	–	–
	LoRA	66	+0.037	74	9	0	–	–	–	–	+0.033	71	9	0	–	–	–	–
	$\text{LoRA}_{\text{late}}$	66	+0.034	65	5	0	-0.003	47	2	2	+0.030	67	9	0	–	–	–	–
	LoRA+ABMIL	66	+0.052	77	12	2	+0.015	71	3	2	+0.048	76	17	0	–	–	–	–
	DoRA	66	+0.037	77	8	0	0.000	50	2	3	+0.033	71	9	0	–	–	–	–
	rsLoRA	66	+0.028	70	9	0	-0.008	35	0	3	+0.024	65	11	0	–	–	–	–
	Lin→LoRA	66	+0.035	71	6	0	-0.001	44	2	3	+0.031	71	8	0	–	–	–	–
	Attn→LoRA	66	+0.016	59	6	2	-0.021	33	0	3	+0.012	56	3	0	–	–	–	–
	Attn-only FT	66	+0.039	70	3	3	+0.002	55	5	5	+0.035	67	6	2	–	–	–	–
	Full FT†	66	-0.117	21	0	32	-0.153	6	0	39	-0.121	14	0	32	–	–	–	–
Pillar-0	Lin	12	–	–	–	–	-0.033	8	0	83	+0.001	58	8	33	-0.004	33	17	50
	Attn	12	-0.001	42	33	8	-0.034	8	0	92	–	–	–	–	-0.005	33	17	58
	LoRA	12	+0.033	92	83	0	–	–	–	–	+0.034	92	92	0	+0.029	100	67	0
	$\text{LoRA}_{\text{late}}$	12	+0.016	75	67	0	-0.017	0	0	83	+0.017	75	67	8	+0.012	58	33	25
	LoRA+ABMIL	12	+0.026	83	75	8	-0.007	50	42	33	+0.028	92	83	0	+0.022	58	58	8
	DoRA	12	+0.034	92	83	0	+0.001	50	0	8	+0.035	100	92	0	+0.030	83	67	0
	rsLoRA	12	+0.032	100	83	0	-0.001	75	17	8	+0.033	100	92	0	+0.028	92	67	0
	Lin→LoRA	12	+0.022	100	83	0	-0.012	8	8	58	+0.023	100	92	0	+0.018	67	42	25
	Attn→LoRA	12	+0.013	92	58	0	-0.020	8	0	83	+0.014	100	58	0	+0.009	50	33	25
	Attn-only FT	12	+0.034	92	83	0	+0.001	75	8	0	+0.036	92	83	0	+0.030	100	83	0
	Full FT	12	+0.004	67	50	17	-0.029	0	0	67	+0.005	67	42	8	–	–	–	–

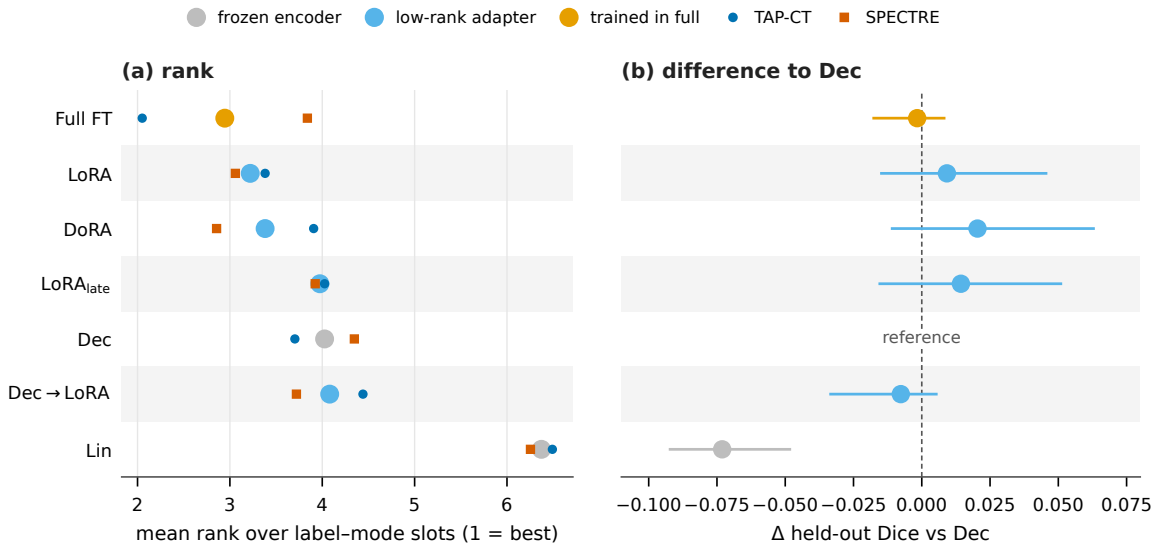


Figure 15: Segmentation methods on TAP-CT and SPECTRE. (a) Mean rank over structure–mode pairs, 1 = best. (b) Label-level mean difference to the frozen decoder with its 95% interval.

Table 22: Segmentation paired tests: share of structure–mode pairs in which the method is ahead of the baseline, and significantly ahead or behind (paired case bootstrap, Benjamini–Hochberg, adjusted $p < 0.05$), per backbone.

Backbone	Method	n	vs Lin				vs Dec				vs LoRA			
			Δ	>0	+	−	Δ	>0	+	−	Δ	>0	+	−
TAP-CT	Lin	74	–	–	–	–	-0.086	4	4	93	-0.088	4	3	93
	Dec	74	+0.086	96	93	4	–	–	–	–	-0.002	43	30	31
	LoRA	74	+0.088	96	93	3	+0.002	54	31	30	–	–	–	–
	LoRA$_{late}$	59	+0.077	95	92	5	-0.006	53	29	32	-0.012	39	14	29
	DoRA	59	+0.078	95	90	5	-0.006	53	25	25	-0.011	34	10	31
	Full FT	74	+0.085	93	93	7	-0.001	76	53	8	-0.002	70	54	12
	Dec→LoRA	59	+0.066	92	88	8	-0.017	29	7	44	-0.024	29	20	37
SPECTRE	Lin	74	–	–	–	–	-0.061	8	5	86	-0.095	7	4	92
	Dec	74	+0.061	92	86	5	–	–	–	–	-0.034	19	8	57
	LoRA	74	+0.095	93	93	4	+0.034	80	57	8	–	–	–	–
	LoRA$_{late}$	59	+0.079	92	90	5	+0.032	69	46	8	-0.009	27	8	42
	DoRA	59	+0.088	93	90	7	+0.041	76	54	12	+0.001	58	25	15
	Full FT	74	+0.059	89	81	8	-0.002	54	38	20	-0.035	36	27	31
	Dec→LoRA	59	+0.051	90	85	7	+0.004	69	20	12	-0.037	25	10	49

Table 23: Segmentation per mode: LoRA − Dec and Full FT − Dec in dataset-mean held-out Dice per backbone and mode, including the 8000-step joint models of AMOS22 and TotalSegmentator.

Dataset (structures)	Backbone	per-task		joint		joint, 8000 steps	
		LoRA − Dec	Full FT − Dec	LoRA − Dec	Full FT − Dec	LoRA − Dec	Full FT − Dec
AMOS22 (15)	TAP-CT	-0.026	+0.012	+0.031	-0.027	-0.125	+0.007
	SPECTRE	+0.004	-0.016	+0.126	+0.011	-0.071	+0.010
KiTS23 (3)	TAP-CT	-0.021	+0.005	+0.068	+0.006	–	–
	SPECTRE	+0.014	-0.071	+0.016	-0.052	–	–
LiTS (2)	TAP-CT	+0.007	+0.001	+0.016	+0.016	–	–
	SPECTRE	+0.015	-0.002	+0.011	-0.030	–	–
MSD Pancreas (2)	TAP-CT	+0.013	+0.021	+0.025	+0.008	–	–
	SPECTRE	+0.047	0.000	+0.037	+0.016	–	–
TotalSegmentator (15)	TAP-CT	-0.024	-0.011	-0.022	+0.017	-0.199	+0.022
	SPECTRE	+0.001	-0.009	-0.036	+0.025	-0.254	+0.037

Table 24: Held-out AUROC as in Table 16 for every backbone, with the 26 000-step CVPR26 joint schedule as an extra row.

Cohort (labels)	Backbone	Mode	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ ABMIL	DoRA	rsLoRA	Lin→ LoRA	Attn→ LoRA	Attn-only FT	Full FT
CVPR26 (17)	TAP-CT	per-task	0.677	0.718	0.732	0.704	0.728	0.717	0.724	0.697	0.707	**0.744**	0.720
		joint	0.707	0.720	0.733	0.726	0.729	0.733	0.722	0.695	0.708	0.739	**0.744**
		joint (26 000 steps)	0.693	0.722	0.692	0.728	0.727	0.720	0.701	0.698	0.726	**0.756**	0.650
	SPECTRE	per-task	0.736	0.725	0.769	0.769	0.776	0.766	0.768	0.769	0.765	**0.778**	0.627†
		joint	0.702	0.738	0.716	0.731	**0.747**	0.731	0.690	0.744	0.728	0.740	0.575†
		joint (26 000 steps)	0.744	0.713	0.755	0.762	0.760	0.752	**0.763**	0.746	0.758	0.756	0.574†
FDG-PET/CT (4)	TAP-CT	per-task	0.863	0.836	0.852	**0.868**	0.839	0.847	0.817	0.866	0.856	0.832	0.769
		joint	0.858	0.860	0.866	0.864	0.849	0.858	0.785	0.860	0.862	**0.870**	0.608
	SPECTRE	per-task	0.857	0.862	0.882	0.879	0.874	**0.892**	0.872	0.887	0.867	0.848	0.617†
		joint	0.884	0.880	0.901	0.888	0.889	0.900	0.900	0.884	0.884	**0.906**	0.673†
	VoCo	per-task	0.720	0.685	0.512	0.671	0.562	0.669	0.598	**0.736**	0.591	0.665	0.606
		joint	0.745	0.739	0.600	0.506	0.655	0.564	0.527	**0.750**	0.520	0.641	0.582
RSNA 2022 C-spine (8)	TAP-CT	per-task	0.619	0.625	0.623	0.608	0.613	0.641	0.635	0.606	0.629	**0.660**	0.637
		joint	0.626	0.624	0.714	0.706	0.713	0.680	0.601	0.624	0.620	0.712	**0.732**
	SPECTRE	per-task	0.579	0.591	0.635	**0.654**	0.644	0.619	0.638	0.615	0.591	0.635	0.566†
		joint	0.629	0.601	0.706	0.644	**0.730**	0.692	0.691	0.670	0.611	0.718	0.554†
	VoCo	per-task	0.606	0.508	0.537	0.551	0.596	0.561	**0.617**	0.602	0.521	0.576	0.605
		joint	0.586	0.548	0.560	0.550	0.565	0.589	0.578	**0.619**	0.527	0.589	0.593
RSNA 2023 trauma (4)	TAP-CT	per-task	0.737	0.718	0.739	0.756	0.703	0.752	0.725	0.738	0.716	**0.859**	0.775
		joint	0.739	0.738	0.757	0.744	0.758	0.768	0.684	0.746	0.758	0.833	**0.881**
	SPECTRE	per-task	0.791	0.778	0.824	0.832	**0.852**	0.842	0.836	0.825	0.798	0.772	0.624†
		joint	0.789	0.788	0.846	0.835	**0.859**	0.838	0.850	0.827	0.810	0.803	0.654†
	VoCo	per-task	0.617	0.625	0.622	0.601	0.595	0.611	0.608	**0.631**	0.569	0.588	0.596
		joint	0.631	0.616	0.603	0.627	0.570	0.583	0.607	0.626	0.635	0.593	**0.648**
RSNA 2019 hemorrhage (6)	TAP-CT	per-task	0.870	0.873	0.894	0.894	0.891	0.897	0.870	0.877	0.878	**0.911**	0.888
		joint	0.871	0.877	0.906	0.895	0.886	0.897	0.869	0.868	0.876	**0.915**	0.854
	Pillar-0	per-task	0.886	0.884	0.929	0.914	0.902	**0.931**	0.920	0.912	0.903	**0.931**	0.904
		joint	0.888	0.888	0.912	0.892	**0.925**	0.912	0.918	0.905	0.898	0.912	0.878

TAP-CT classification.

CVPR26, TAP-CT: held-out AUROC per label

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ ABMIL	DoRA	rsLoRA	Lin→ LoRA	Attn→ LoRA	Attn-only FT	Full FT
per-task											
adrenal hyperplasia (AMOS)	0.797	0.836	0.803	0.788	0.810	0.771	0.789	0.805	0.820	0.840	**0.874**
ascites (AMOS)	0.776	**0.835**	0.767	0.737	0.784	0.710	0.775	0.763	0.753	0.682	0.820
atherosclerosis (AMOS)	0.754	0.871	**0.890**	0.835	0.829	0.838	0.857	0.750	0.862	0.814	0.877
cholecystitis (AMOS)	0.730	**0.782**	0.744	0.689	0.735	0.749	0.737	0.747	0.718	0.770	0.738
colorectal cancer (AMOS)	0.562	0.652	0.649	0.649	0.670	0.628	0.622	0.604	0.637	**0.699**	0.616
fatty liver (AMOS)	0.872	0.923	0.950	0.947	0.953	0.938	0.992	0.903	0.917	**0.994**	0.981
gallstone (AMOS)	0.601	0.650	0.662	0.642	0.682	0.676	0.605	0.609	**0.693**	0.647	0.690
hydronephrosis (AMOS)	0.863	0.821	0.896	0.858	0.858	0.846	0.775	0.867	**0.904**	0.856	0.669
kidney stone (AMOS)	0.625	0.692	0.628	0.650	0.640	0.619	0.700	0.695	**0.715**	0.700	0.673
liver calcifications (AMOS)	0.441	0.372	0.492	0.438	0.469	0.508	0.506	0.430	0.389	**0.576**	0.574
liver cyst (AMOS)	0.641	0.673	0.669	0.644	0.648	0.625	0.677	0.663	0.643	**0.692**	0.657
liver lesion (AMOS)	0.569	0.639	**0.706**	0.691	0.608	0.661	0.649	0.587	0.642	0.657	0.696
lymphadenopathy (AMOS)	0.600	0.733	0.643	0.669	**0.765**	0.733	0.657	0.693	0.695	0.760	0.543
renal cyst (AMOS)	0.659	0.637	0.671	**0.691**	0.654	0.643	0.686	0.641	0.639	0.659	0.666
splenomegaly (AMOS)	0.684	**0.751**	0.707	0.724	0.714	0.708	0.733	0.702	0.714	0.718	0.709
covid (COVID-CT)	0.814	0.758	0.971	0.750	0.992	0.988	0.980	0.840	0.770	**0.996**	0.916
lung nodule malignancy (LUNA25)	0.527	0.572	**0.593**	0.563	0.565	0.550	0.574	0.558	0.505	0.590	0.546
joint											
adrenal hyperplasia (AMOS)	0.840	0.835	0.846	0.838	0.868	0.820	0.820	0.872	0.833	0.803	**0.874**
ascites (AMOS)	0.769	0.804	0.839	0.796	0.747	0.814	0.808	0.767	0.678	**0.890**	0.810
atherosclerosis (AMOS)	0.835	0.805	**0.869**	0.835	0.814	0.812	0.849	0.816	0.779	0.831	0.842
cholecystitis (AMOS)	0.756	0.704	0.706	**0.771**	0.759	0.728	0.727	0.713	0.709	0.721	0.720
colorectal cancer (AMOS)	0.619	0.637	**0.696**	0.670	0.667	0.661	0.667	0.607	0.679	0.670	0.637
fatty liver (AMOS)	0.844	0.795	0.777	0.758	0.820	0.787	0.775	0.822	0.755	0.811	**0.878**
gallstone (AMOS)	0.675	0.663	0.647	0.633	0.661	0.675	0.664	0.666	0.662	**0.679**	0.672
hydronephrosis (AMOS)	**0.860**	0.829	0.787	0.769	0.810	0.819	0.704	0.831	0.787	0.775	0.758
kidney stone (AMOS)	**0.738**	0.685	0.702	0.710	0.710	0.675	0.631	0.685	0.684	0.633	0.715
liver calcifications (AMOS)	0.435	0.512	0.460	0.512	0.503	0.523	0.466	0.398	0.512	0.545	**0.579**
liver cyst (AMOS)	0.632	0.667	0.652	0.660	0.656	0.659	0.647	0.633	0.630	**0.673**	0.667
liver lesion (AMOS)	0.631	**0.714**	0.696	0.689	0.703	0.684	0.697	0.664	0.696	0.687	0.659
lymphadenopathy (AMOS)	0.727	0.766	**0.781**	0.735	0.708	0.747	0.777	0.733	0.680	0.735	0.777
renal cyst (AMOS)	0.597	0.637	0.665	0.660	0.684	0.684	0.657	0.614	0.647	**0.699**	0.671
splenomegaly (AMOS)	0.715	0.784	0.860	0.815	0.798	**0.909**	0.872	0.683	0.784	0.862	0.836
covid (COVID-CT)	0.828	0.738	0.832	0.820	0.844	0.820	0.855	0.824	0.842	0.895	**0.934**
lung nodule malignancy (LUNA25)	0.514	0.667	0.645	0.672	0.640	0.646	0.654	0.479	**0.682**	0.655	0.620
joint (26 000 steps)											
adrenal hyperplasia (AMOS)	0.797	0.763	0.617	0.795	0.763	0.812	0.850	0.814	0.803	**0.883**	**0.883**
ascites (AMOS)	0.790	0.806	0.824	0.847	0.767	**0.865**	0.816	0.778	0.716	0.763	0.620
atherosclerosis (AMOS)	0.800	0.820	0.842	0.824	0.739	0.774	0.800	0.790	0.781	**0.851**	0.621
cholecystitis (AMOS)	0.807	0.705	0.686	0.713	0.735	0.706	0.678	**0.867**	0.839	0.726	0.659
colorectal cancer (AMOS)	0.601	0.708	0.589	0.696	0.613	0.631	0.708	0.583	**0.717**	0.685	0.580
fatty liver (AMOS)	0.853	0.853	0.789	0.847	0.884	0.802	0.723	0.856	0.853	**0.897**	0.731
gallstone (AMOS)	0.624	0.616	0.660	0.624	0.652	0.678	**0.683**	0.651	0.631	0.664	0.652
hydronephrosis (AMOS)	0.833	**0.838**	0.765	0.738	0.792	0.765	0.688	**0.838**	0.804	0.773	0.696
kidney stone (AMOS)	0.679	0.722	0.652	0.672	0.673	0.653	0.655	0.687	**0.727**	0.668	0.590
liver calcifications (AMOS)	0.486	0.430	0.443	**0.539**	0.437	0.515	0.478	0.392	0.424	0.525	0.399
liver cyst (AMOS)	0.645	0.686	0.630	0.706	0.719	**0.720**	0.626	0.627	0.676	0.668	0.586
liver lesion (AMOS)	0.540	**0.714**	0.702	0.659	0.701	0.701	0.655	0.605	0.708	0.703	0.672
lymphadenopathy (AMOS)	0.688	0.726	0.632	0.741	**0.778**	0.776	0.683	0.693	0.737	0.770	0.670
renal cyst (AMOS)	0.611	0.664	0.633	0.686	**0.721**	0.658	0.682	0.622	0.655	0.684	0.576
splenomegaly (AMOS)	0.733	0.826	0.871	0.807	0.805	**0.879**	0.750	0.727	0.791	0.817	0.760
covid (COVID-CT)	0.781	0.762	0.809	0.801	0.938	0.777	0.852	0.809	0.822	**0.992**	0.795
lung nodule malignancy (LUNA25)	0.516	0.641	0.630	0.680	0.647	0.529	0.583	0.526	0.649	**0.784**	0.564

FDG-PET/CT, TAP-CT: held-out AUROC per label

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ ABMIL	DoRA	rsLoRA	Lin→ LoRA	Attn→ LoRA	Attn-only FT	Full FT
per-task											
negative	0.780	0.774	0.766	**0.815**	0.768	0.733	0.669	0.774	0.748	0.681	0.695
melanoma	0.813	0.758	0.794	0.829	0.808	0.818	0.763	**0.835**	0.810	0.771	0.724
lung cancer	0.925	0.927	0.919	0.933	0.855	0.901	0.901	0.923	0.908	**0.936**	0.856
lymphoma	0.933	0.886	0.930	0.895	0.924	0.935	0.936	0.931	**0.960**	0.941	0.800
joint											
negative	0.763	**0.811**	0.807	0.764	0.762	0.763	0.650	0.767	0.788	0.797	0.576
melanoma	0.798	0.787	0.792	0.796	**0.813**	0.809	0.776	0.808	0.776	0.795	0.709
lung cancer	0.933	0.920	0.944	**0.945**	0.894	0.927	0.810	0.931	0.921	0.934	0.568
lymphoma	0.937	0.925	0.919	0.951	0.925	0.931	0.907	0.933	**0.963**	0.953	0.580

RSNA 2022 C-spine, TAP-CT: held-out AUROC per label

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ ABMIL	DoRA	rsLoRA	Lin→ LoRA	Attn→ LoRA	Attn-only FT	Full FT
per-task											
patient overall	0.649	0.683	0.682	0.673	0.748	0.764	0.682	0.682	0.669	**0.781**	0.623
C1	0.673	0.714	0.681	0.725	0.699	0.717	0.737	0.686	0.715	**0.774**	0.699
C2	0.708	0.759	0.732	0.755	0.749	0.743	0.723	0.704	0.730	**0.766**	0.696
C3	**0.635**	0.555	0.401	0.397	0.465	0.547	0.540	0.511	0.529	0.592	0.531
C4	0.643	0.576	0.562	0.621	0.508	0.530	0.514	0.637	0.562	0.522	**0.655**
C5	0.515	0.529	0.589	0.496	0.443	0.515	0.597	0.494	**0.601**	0.537	0.567
C6	0.496	0.619	0.658	0.586	0.664	0.648	0.639	0.502	0.641	0.654	**0.680**
C7	0.632	0.567	**0.679**	0.611	0.629	0.661	0.649	0.629	0.586	0.655	0.646
joint											
patient overall	0.672	0.653	0.750	0.722	0.734	0.701	0.593	0.677	0.667	0.753	**0.769**
C1	0.641	0.707	0.721	0.750	0.716	0.711	0.681	0.622	0.713	0.704	**0.758**
C2	0.760	0.727	0.764	0.751	0.766	0.738	0.702	0.744	0.724	0.787	**0.810**
C3	0.550	0.455	0.631	**0.692**	**0.692**	0.591	0.470	0.590	0.426	0.631	0.624
C4	0.690	0.652	0.741	0.707	0.697	0.712	0.529	0.672	0.648	0.732	**0.782**
C5	0.519	0.558	0.686	0.646	0.683	0.627	0.565	0.507	0.566	0.692	**0.693**
C6	0.540	0.581	0.708	0.684	**0.715**	0.684	0.689	0.534	0.574	0.680	0.694
C7	0.635	0.658	0.713	0.697	0.698	0.679	0.582	0.649	0.641	0.719	**0.727**

RSNA 2023 trauma, TAP-CT: held-out AUROC per label

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ ABMIL	DoRA	rsLoRA	Lin→ LoRA	Attn→ LoRA	Attn-only FT	Full FT
per-task											
any injury	0.751	0.777	0.770	0.762	0.779	0.769	0.706	0.759	0.748	**0.880**	0.866
kidney injury	0.704	0.636	0.712	0.688	0.654	0.797	0.862	0.674	0.686	**0.876**	0.792
liver injury	0.777	0.768	0.758	**0.799**	0.777	0.783	0.738	0.769	0.793	0.789	0.688
spleen injury	0.717	0.691	0.714	0.774	0.604	0.660	0.593	0.750	0.637	**0.893**	0.752
joint											
any injury	0.755	0.758	0.770	0.754	0.763	0.789	0.693	0.762	0.769	0.842	**0.874**
kidney injury	0.716	0.697	0.738	0.710	0.776	0.752	0.709	0.698	0.755	0.841	**0.963**
liver injury	0.800	0.786	0.815	0.808	0.811	0.790	0.718	0.805	0.795	**0.844**	0.831
spleen injury	0.685	0.711	0.706	0.707	0.681	0.740	0.616	0.719	0.713	0.807	**0.854**

RSNA 2019 hemorrhage, TAP-CT: held-out AUROC per label

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ ABMIL	DoRA	rsLoRA	Lin→ LoRA	Attn→ LoRA	Attn-only FT	Full FT
per-task											
any	0.916	0.919	**0.947**	0.940	0.933	0.938	0.919	0.922	0.923	0.946	0.942
epidural	0.812	0.803	0.781	0.790	0.776	0.819	0.764	0.820	0.788	0.839	**0.866**
intraparenchymal	0.875	0.880	0.926	0.926	0.916	0.923	0.921	0.888	0.902	**0.938**	0.883
intraventricular	0.928	0.937	0.945	0.948	0.951	0.951	0.930	0.934	0.944	**0.964**	0.949
subarachnoid	0.852	0.857	0.877	**0.885**	**0.885**	0.880	0.857	0.856	0.861	0.870	0.875
subdural	0.838	0.842	0.887	0.874	0.883	0.871	0.832	0.843	0.849	**0.907**	0.814
joint											
any	0.915	0.918	0.947	0.938	0.937	0.937	0.926	0.914	0.920	**0.953**	0.911
epidural	0.814	0.831	0.877	0.860	0.824	0.874	0.792	0.804	0.820	**0.890**	0.747
intraparenchymal	0.875	0.881	0.919	0.908	0.911	0.913	0.900	0.875	0.884	**0.925**	0.907
intraventricular	0.926	0.933	0.947	0.944	0.935	0.943	0.921	0.924	0.933	**0.949**	0.927
subarachnoid	0.852	0.862	0.877	0.869	0.859	0.868	0.866	0.853	0.862	**0.878**	0.858
subdural	0.841	0.840	0.869	0.851	0.848	0.847	0.808	0.838	0.834	**0.893**	0.776

SPECTRE classification.

CVPR26, SPECTRE: held-out AUROC per label

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ ABMIL	DoRA	rsLoRA	Lin→ LoRA	Attn→ LoRA	Attn-only FT	Full FT
per-task											
adrenal hyperplasia (AMOS)	0.806	0.523	0.835	0.803	**0.859**	0.586	0.756	0.805	0.759	0.780	0.763†
ascites (AMOS)	0.894	0.898	**0.922**	0.876	0.876	0.902	0.878	**0.922**	0.886	0.798	0.545†
atherosclerosis (AMOS)	0.761	0.785	0.801	0.831	0.840	**0.844**	0.792	0.744	0.700	0.779	0.818†
cholecystitis (AMOS)	0.735	0.725	0.804	0.801	**0.884**	0.838	0.817	0.826	0.791	0.804	0.650†
colorectal cancer (AMOS)	0.604	0.696	0.714	0.655	0.557	0.720	0.667	0.744	**0.747**	0.738	0.500†
fatty liver (AMOS)	0.808	0.787	0.947	0.966	**0.981**	0.916	0.947	0.903	0.944	0.944	0.767†
gallstone (AMOS)	0.658	0.700	0.641	0.692	0.794	0.692	0.775	0.644	0.747	**0.825**	0.571†
hydronephrosis (AMOS)	0.850	0.854	0.908	0.925	**0.942**	0.902	0.869	0.931	0.904	0.940	0.556†
kidney stone (AMOS)	0.662	0.684	0.712	0.741	0.722	0.672	0.694	0.668	0.680	**0.761**	0.591†
liver calcifications (AMOS)	0.582	**0.693**	0.635	0.515	0.522	0.628	0.557	0.622	0.618	0.568	0.550†
liver cyst (AMOS)	0.580	0.631	0.646	0.652	0.649	0.655	0.644	0.655	0.662	**0.663**	0.623†
liver lesion (AMOS)	0.683	0.641	0.619	0.709	0.703	0.701	**0.753**	0.694	0.689	0.722	0.605†
lymphadenopathy (AMOS)	0.758	0.607	0.781	0.735	0.771	**0.828**	0.781	0.787	0.727	0.697	0.526†
renal cyst (AMOS)	0.628	0.650	0.573	**0.716**	0.707	0.576	0.568	0.602	0.667	0.703	0.476†
splenomegaly (AMOS)	0.759	0.704	**0.872**	0.765	0.698	0.868	0.828	0.821	0.765	0.762	0.574†
covid (COVID-CT)	**1.000**	0.996	0.988	0.980	0.932	0.984	0.988	0.996	0.996	0.984	0.723†
lung nodule malignancy (LUNA25)	0.737	0.743	0.679	0.705	0.751	0.711	0.736	0.717	0.729	**0.759**	0.820†
joint											
adrenal hyperplasia (AMOS)	0.724	0.566	0.523	0.628	0.665	0.538	0.690	**0.780**	0.647	0.737	0.500†
ascites (AMOS)	0.825	**0.871**	0.839	0.810	0.835	0.863	0.812	0.784	0.816	0.867	0.596†
atherosclerosis (AMOS)	0.783	0.759	0.761	0.776	0.794	0.743	0.719	**0.812**	0.770	0.743	0.529†
cholecystitis (AMOS)	0.683	**0.796**	0.757	0.676	0.773	0.729	0.708	0.712	0.692	0.781	0.660†
colorectal cancer (AMOS)	0.693	0.708	0.667	0.670	0.705	0.714	0.658	**0.720**	0.702	0.646	0.345†
fatty liver (AMOS)	**0.828**	0.714	0.691	0.770	0.800	0.752	0.591	0.797	0.695	0.805	0.667†
gallstone (AMOS)	0.476	0.751	0.675	0.680	0.772	0.773	0.754	0.752	**0.797**	0.693	0.657†
hydronephrosis (AMOS)	**0.900**	0.812	0.760	0.779	0.738	0.783	0.754	0.863	0.692	0.821	0.742†
kidney stone (AMOS)	0.629	0.665	0.658	**0.702**	0.696	0.662	0.665	0.607	0.701	0.684	0.529†
liver calcifications (AMOS)	0.336	0.704	0.633	0.680	0.646	0.560	0.471	0.545	**0.734**	0.491	0.497†
liver cyst (AMOS)	0.619	0.646	0.657	0.651	0.685	**0.712**	0.619	0.651	0.633	0.692	0.550†
liver lesion (AMOS)	0.615	0.628	0.637	0.611	0.648	0.634	**0.683**	0.662	0.594	0.649	0.516†
lymphadenopathy (AMOS)	0.801	0.778	**0.807**	0.751	0.787	0.776	0.730	0.801	0.742	0.777	0.500†
renal cyst (AMOS)	0.635	0.627	0.631	**0.695**	0.643	0.682	0.616	0.671	0.603	0.673	0.520†
splenomegaly (AMOS)	0.655	0.785	0.791	**0.846**	0.821	0.822	0.613	0.761	**0.846**	0.826	0.716†
covid (COVID-CT)	0.988	**1.000**	0.986	0.996	0.996	0.996	0.996	0.992	**1.000**	0.984	0.713†
lung nodule malignancy (LUNA25)	**0.736**	0.728	0.706	0.713	0.702	0.685	0.648	**0.736**	0.719	0.720	0.541†
joint (26 000 steps)											
adrenal hyperplasia (AMOS)	0.731	0.444	0.699	0.718	0.712	0.639	0.652	0.703	0.667	**0.742**	0.288†
ascites (AMOS)	0.884	0.892	0.853	0.882	**0.894**	0.873	0.875	0.847	0.861	0.869	0.486†
atherosclerosis (AMOS)	0.829	0.735	0.824	0.787	0.724	0.767	0.812	0.792	**0.846**	0.809	0.665†
cholecystitis (AMOS)	0.760	0.705	0.769	**0.782**	0.763	0.714	0.763	0.620	0.723	0.657	0.471†
colorectal cancer (AMOS)	0.714	0.583	0.637	0.738	0.729	0.711	0.768	**0.810**	0.768	0.673	0.699†
fatty liver (AMOS)	0.822	0.644	0.897	**0.922**	0.916	0.909	0.877	0.844	0.848	0.841	0.787†
gallstone (AMOS)	0.702	**0.784**	0.656	0.687	0.769	0.669	0.664	0.685	0.692	0.705	0.598†
hydronephrosis (AMOS)	0.815	0.773	0.842	0.881	0.833	0.779	0.846	0.831	**0.887**	0.825	0.562†
kidney stone (AMOS)	0.612	0.682	0.705	0.686	0.697	0.734	**0.743**	0.716	0.726	0.690	0.542†
liver calcifications (AMOS)	0.514	**0.737**	0.545	0.509	0.515	0.599	0.560	0.543	0.635	0.588	0.522†
liver cyst (AMOS)	0.644	0.637	**0.718**	0.706	0.689	0.704	0.701	0.672	0.675	0.682	0.427†
liver lesion (AMOS)	0.618	0.621	0.675	0.669	0.672	0.660	0.664	0.657	0.620	**0.679**	0.617†
lymphadenopathy (AMOS)	0.762	0.734	**0.821**	0.747	0.751	0.803	0.794	0.762	0.781	0.798	0.468†
renal cyst (AMOS)	0.613	0.636	0.654	0.690	0.675	**0.703**	0.683	0.600	0.659	0.684	0.519†
splenomegaly (AMOS)	0.836	0.780	0.871	0.822	0.824	0.832	**0.884**	0.828	0.803	0.865	0.501†
covid (COVID-CT)	0.996	0.992	0.996	0.992	0.992	0.988	0.988	**1.000**	0.996	0.984	0.945†
lung nodule malignancy (LUNA25)	**0.794**	0.741	0.678	0.737	0.766	0.702	0.704	0.768	0.693	0.755	0.667†

FDG-PET/CT, SPECTRE: held-out AUROC per label

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ ABMIL	DoRA	rsLoRA	Lin→ LoRA	Attn→ LoRA	Attn-only FT	Full FT
per-task											
negative	0.841	0.809	0.832	0.806	0.834	0.843	**0.852**	0.843	0.794	0.647	0.533†
melanoma	0.725	0.775	0.787	0.779	0.736	0.795	0.750	0.795	0.766	**0.806**	0.634†
lung cancer	0.926	0.941	0.951	0.963	0.959	0.962	0.951	0.962	0.956	**0.973**	0.801†
lymphoma	0.936	0.922	0.959	0.967	0.965	**0.968**	0.937	0.947	0.951	0.966	0.500†
joint											
negative	0.832	0.837	0.834	0.825	0.822	0.830	0.837	0.815	0.841	**0.846**	0.636†
melanoma	0.804	0.766	0.819	0.783	0.797	0.824	0.813	0.816	0.774	**0.846**	0.604†
lung cancer	0.974	0.966	**0.988**	0.981	0.985	**0.988**	0.987	0.958	0.979	0.986	0.756†
lymphoma	0.926	0.950	0.961	0.963	0.952	0.959	**0.965**	0.949	0.942	0.944	0.695†

RSNA 2022 C-spine, SPECTRE: held-out AUROC per label

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ ABMIL	DoRA	rsLoRA	Lin→ LoRA	Attn→ LoRA	Attn-only FT	Full FT
per-task											
patient overall	0.599	0.643	0.716	0.683	0.722	0.659	0.700	0.684	0.707	**0.778**	0.584†
C1	0.546	0.565	**0.684**	0.640	0.611	0.587	0.590	0.657	0.609	0.537	0.576†
C2	0.674	0.617	0.691	0.729	0.769	0.672	**0.770**	0.740	0.637	0.664	0.685†
C3	0.506	0.721	0.768	**0.816**	0.603	0.706	0.630	0.566	0.426	0.712	0.564†
C4	**0.645**	0.349	0.447	0.543	0.610	0.444	0.635	0.468	0.579	0.471	0.476†
C5	0.557	**0.603**	0.459	0.586	0.508	0.540	0.490	0.529	0.494	0.558	0.546†
C6	0.552	0.612	0.665	0.576	0.665	0.677	0.615	0.632	0.609	**0.701**	0.579†
C7	0.550	0.622	0.652	0.657	0.662	0.668	**0.676**	0.642	0.669	0.657	0.519†
joint											
patient overall	0.585	0.633	0.738	0.674	0.744	0.714	0.723	0.681	0.663	**0.753**	0.590†
C1	0.650	0.528	0.733	0.696	0.716	0.720	0.724	0.699	0.688	**0.735**	0.635†
C2	0.668	0.667	0.772	0.749	**0.784**	0.751	0.765	0.768	0.741	0.755	0.681†
C3	0.660	0.562	0.645	0.514	0.679	0.629	0.627	**0.725**	0.516	0.587	0.460†
C4	0.628	0.615	0.746	0.679	**0.808**	0.737	0.700	0.627	0.581	0.748	0.609†
C5	0.599	0.566	0.616	0.553	0.687	0.631	0.588	0.587	0.473	**0.704**	0.458†
C6	0.636	0.590	0.682	0.626	0.683	0.668	0.689	0.635	0.609	**0.736**	0.482†
C7	0.604	0.650	0.715	0.659	**0.735**	0.687	0.711	0.640	0.618	0.725	0.519†

RSNA 2023 trauma, SPECTRE: held-out AUROC per label

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ ABMIL	DoRA	rsLoRA	Lin→ LoRA	Attn→ LoRA	Attn-only FT	Full FT
per-task											
any injury	0.804	0.793	0.862	0.861	**0.874**	0.848	0.859	0.838	0.850	0.854	0.574†
kidney injury	0.849	0.826	0.850	0.901	**0.911**	0.896	0.857	0.867	0.791	0.738	0.630†
liver injury	0.732	0.732	0.769	0.753	0.774	0.770	**0.776**	0.767	0.768	0.696	0.737†
spleen injury	0.780	0.762	0.816	0.814	0.849	**0.853**	0.852	0.829	0.783	0.802	0.556†
joint											
any injury	0.797	0.794	0.865	0.854	**0.871**	0.858	**0.871**	0.848	0.825	0.817	0.650†
kidney injury	0.838	0.825	0.877	0.877	**0.908**	0.857	0.860	0.870	0.850	0.872	0.644†
liver injury	0.726	0.743	0.797	0.790	0.813	0.803	**0.817**	0.777	0.778	0.742	0.735†
spleen injury	0.795	0.789	0.843	0.818	0.845	0.836	**0.851**	0.811	0.786	0.783	0.588†

Pillar-0 classification.

RSNA 2019 hemorrhage, Pillar-0: held-out AUROC per label

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ ABMIL	DoRA	rsLoRA	Lin→ LoRA	Attn→ LoRA	Attn-only FT	Full FT
per-task											
any	0.910	0.928	0.957	0.944	0.952	0.954	**0.958**	0.944	0.939	**0.958**	0.955
epidural	0.816	0.770	0.875	0.857	0.810	**0.892**	0.818	0.856	0.829	0.887	0.825
intraparenchymal	0.910	0.918	0.947	0.933	0.931	0.945	0.947	0.930	0.924	**0.948**	0.942
intraventricular	0.945	0.941	0.962	0.954	0.959	0.956	**0.963**	0.951	0.952	**0.963**	0.961
subarachnoid	0.863	0.862	0.913	0.887	0.845	**0.918**	**0.918**	0.884	0.875	0.916	0.852
subdural	0.872	0.883	**0.919**	0.911	0.917	0.918	0.917	0.908	0.901	0.914	0.890
joint											
any	0.912	0.917	0.951	0.943	0.952	0.953	0.954	0.940	0.933	**0.957**	0.939
epidural	0.826	0.820	0.857	0.815	**0.885**	0.858	0.872	0.846	0.843	0.851	0.782
intraparenchymal	0.911	0.907	0.935	0.923	**0.942**	0.931	0.934	0.923	0.915	0.936	0.924
intraventricular	0.946	0.940	0.940	0.939	**0.952**	0.941	0.948	0.947	0.942	0.937	0.929
subarachnoid	0.863	0.862	0.884	0.865	**0.904**	0.888	0.896	0.880	0.866	0.886	0.875
subdural	0.872	0.885	0.905	0.866	**0.915**	0.903	0.907	0.897	0.886	0.907	0.821

VoCo classification.

FDG-PET/CT, VoCo: held-out AUROC per label

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ ABMIL	DoRA	rsLoRA	Lin→ LoRA	Attn→ LoRA	Attn-only FT	Full FT
per-task											
negative	0.713	0.626	0.561	0.682	0.511	0.563	0.549	**0.731**	0.622	0.601	0.555
melanoma	0.555	0.481	0.432	**0.660**	0.593	0.644	0.514	0.580	0.532	0.544	0.586
lung cancer	**0.801**	0.777	0.469	0.696	0.671	0.714	0.742	0.794	0.664	0.662	0.643
lymphoma	0.811	**0.858**	0.586	0.645	0.471	0.755	0.586	0.840	0.546	0.852	0.639
joint											
negative	**0.736**	0.723	0.591	0.488	0.486	0.539	0.519	0.735	0.521	0.648	0.521
melanoma	0.633	0.691	0.493	0.410	**0.722**	0.431	0.394	0.624	0.486	0.516	0.591
lung cancer	**0.795**	0.731	0.601	0.606	0.669	0.559	0.596	0.791	0.548	0.658	0.636
lymphoma	0.814	0.810	0.717	0.519	0.744	0.725	0.599	**0.850**	0.524	0.739	0.579

RSNA 2022 C-spine, VoCo: held-out AUROC per label

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ ABMIL	DoRA	rsLoRA	Lin→ LoRA	Attn→ LoRA	Attn-only FT	Full FT
per-task											
patient overall	0.617	0.542	0.567	0.611	0.595	0.625	0.597	**0.649**	0.425	0.560	0.544
C1	0.693	0.566	**0.771**	0.623	0.629	0.635	0.765	0.634	0.519	0.652	0.708
C2	0.718	0.642	0.652	**0.764**	0.700	0.756	0.751	0.723	0.641	0.642	0.736
C3	0.500	0.332	0.249	0.322	**0.768**	0.401	0.524	0.496	0.476	0.542	0.439
C4	0.520	0.484	0.519	0.349	0.534	0.404	0.600	0.616	0.610	0.557	**0.727**
C5	0.587	0.572	0.478	0.538	0.473	0.540	0.561	0.492	0.591	0.597	**0.621**
C6	0.604	0.412	0.452	0.567	0.551	0.519	**0.626**	0.589	0.398	0.493	0.489
C7	0.613	0.514	0.605	**0.631**	0.514	0.609	0.509	0.614	0.511	0.562	0.572
joint											
patient overall	0.653	0.521	0.534	0.581	0.563	0.592	0.573	**0.661**	0.496	0.613	0.537
C1	0.670	0.700	0.737	**0.753**	0.741	0.745	0.684	0.675	0.666	0.745	0.726
C2	0.724	0.648	0.716	0.713	0.731	0.725	0.715	0.707	0.618	**0.732**	0.683
C3	0.465	0.504	0.290	0.313	0.493	0.432	0.509	**0.610**	0.482	0.467	0.564
C4	0.548	**0.594**	0.591	0.582	0.568	0.520	0.507	0.590	0.588	0.577	0.551
C5	0.526	0.506	0.472	0.536	0.458	**0.638**	0.530	0.521	0.515	0.475	0.600
C6	0.483	0.434	0.532	0.407	0.418	0.426	0.499	**0.568**	0.406	0.501	0.513
C7	0.620	0.477	0.606	0.520	0.548	**0.631**	0.610	0.621	0.447	0.601	0.573

RSNA 2023 trauma, VoCo: held-out AUROC per label

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ ABMIL	DoRA	rsLoRA	Lin→ LoRA	Attn→ LoRA	Attn-only FT	Full FT
per-task											
any injury	0.621	0.614	0.613	**0.628**	0.593	0.604	0.618	0.626	0.605	0.596	0.611
kidney injury	0.672	0.664	**0.674**	0.538	0.637	0.632	0.637	0.664	0.562	0.568	0.584
liver injury	0.595	**0.637**	0.598	0.619	0.588	0.608	0.593	0.632	0.575	0.599	0.613
spleen injury	0.581	0.584	0.604	**0.619**	0.564	0.601	0.585	0.601	0.532	0.589	0.575
joint											
any injury	0.633	0.614	0.594	**0.635**	0.600	0.625	0.584	0.610	0.632	0.628	0.624
kidney injury	0.673	0.663	0.662	0.668	0.521	0.536	0.643	0.668	0.676	0.536	**0.716**
liver injury	0.619	0.645	0.590	0.640	0.569	0.588	0.625	0.623	0.661	0.590	**0.671**
spleen injury	0.599	0.541	0.566	0.565	0.590	0.582	0.575	0.603	0.572	**0.620**	0.582

TAP-CT segmentation.

AMOS22, TAP-CT: held-out Dice per structure

Structure	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→ LoRA
per-task							
spleen	0.815	0.900	**0.925**	0.917	0.921	0.908	0.896
right kidney	0.704	0.897	**0.922**	0.905	**0.922**	**0.922**	0.895
left kidney	0.721	0.882	0.914	0.912	0.910	**0.918**	0.887
gallbladder	0.546	0.660	0.673	0.658	0.648	**0.692**	0.678
esophagus	0.547	0.741	0.646	0.642	0.650	**0.774**	0.732
liver	0.893	0.951	0.951	**0.954**	0.953	0.951	0.947
stomach	0.778	0.869	0.882	**0.883**	0.874	0.873	0.868
aorta	0.791	0.918	0.892	0.893	0.889	**0.921**	0.915
postcava	0.698	0.824	0.772	0.772	0.767	**0.836**	0.819
pancreas	0.676	**0.780**	0.776	0.776	0.775	0.778	0.776
right adrenal	0.331	0.563	0.415	0.447	0.415	**0.575**	0.563
left adrenal	0.378	0.580	0.421	0.411	0.407	**0.586**	0.574
duodenum	0.532	0.677	0.676	0.668	0.654	**0.697**	0.671
bladder	0.736	0.850	0.856	0.856	0.857	**0.863**	0.848
prostate uterus	0.726	**0.798**	0.779	0.764	0.768	0.783	0.792
joint							
spleen	0.808	0.908	0.918	0.916	0.921	**0.925**	0.904
right kidney	0.679	0.891	0.916	0.913	0.914	**0.927**	0.890
left kidney	0.707	0.880	0.912	0.914	0.913	**0.925**	0.881
gallbladder	0.497	0.665	0.650	0.659	0.686	**0.696**	0.652
esophagus	0.524	0.713	**0.728**	0.656	0.714	0.000	0.000
liver	0.893	0.943	0.934	0.947	0.938	**0.953**	0.941
stomach	0.776	0.875	0.874	0.854	0.868	**0.877**	0.868
aorta	0.782	0.904	0.910	0.900	0.908	**0.922**	0.902
postcava	0.695	0.822	0.827	0.757	0.824	**0.844**	0.817
pancreas	0.665	0.773	0.773	0.757	0.772	**0.787**	0.766
right adrenal	0.337	0.000	**0.566**	0.000	0.000	0.000	0.000
left adrenal	**0.361**	0.000	0.000	0.000	0.000	0.000	0.000
duodenum	0.516	0.677	0.681	0.666	0.686	**0.716**	0.679
bladder	0.729	0.806	0.675	0.758	0.750	**0.855**	0.777
prostate uterus	0.718	0.766	0.715	0.770	0.718	**0.789**	0.763
joint (8000 steps)							
spleen	0.796	0.898	0.761	0.742	0.711	**0.916**	0.898
right kidney	0.665	0.882	0.774	0.666	0.874	**0.909**	0.872
left kidney	0.686	0.871	0.884	0.903	0.884	**0.910**	0.866
gallbladder	0.470	0.626	0.000	0.000	0.000	**0.668**	0.621
esophagus	0.512	0.000	0.000	0.000	0.000	**0.675**	0.000
liver	0.888	0.937	0.946	**0.947**	0.946	0.945	0.936
stomach	0.766	0.852	**0.864**	0.811	0.802	0.862	0.841
aorta	0.782	0.896	0.864	0.722	0.789	**0.900**	0.890
postcava	0.691	0.799	0.000	0.000	0.000	**0.816**	0.791
pancreas	0.657	0.746	0.639	0.397	0.630	**0.760**	0.734
right adrenal	**0.323**	0.000	0.000	0.000	0.000	0.000	0.000
left adrenal	**0.356**	0.000	0.000	0.000	0.000	0.000	0.000
duodenum	0.501	0.654	0.628	0.529	0.502	**0.676**	0.639
bladder	0.698	0.763	0.000	0.000	0.000	0.000	**0.772**
prostate uterus	**0.695**	0.000	0.687	0.458	0.682	0.000	0.000

KiTS23, TAP-CT: held-out Dice per structure

Structure	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→ LoRA
per-task							
kidney	0.887	0.937	0.940	**0.942**	**0.942**	0.936	0.937
tumor	0.536	0.625	0.571	0.594	0.580	0.628	**0.632**
cyst	0.171	0.189	0.178	0.166	0.167	**0.201**	0.191
joint							
kidney	0.889	0.910	**0.937**	0.936	0.935	0.913	0.925
tumor	0.360	0.544	**0.624**	0.592	0.600	0.561	0.574
cyst	0.040	0.087	**0.185**	0.140	0.146	0.084	0.000

LiTS, TAP-CT: held-out Dice per structure

Structure	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→ LoRA
per-task							
liver	0.916	0.951	0.960	0.959	0.961	**0.963**	0.957
tumor	0.403	0.468	**0.473**	0.459	0.467	0.457	0.471
joint							
liver	0.913	0.950	0.958	0.957	0.957	**0.961**	0.952
tumor	0.399	0.448	0.472	**0.485**	0.427	0.469	0.454

MSD Pancreas, TAP-CT: held-out Dice per structure

Structure	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→ LoRA
per-task							
pancreas	0.663	0.751	**0.766**	0.761	0.757	0.760	0.748
tumor	0.190	0.231	0.242	0.241	0.247	**0.263**	0.241
joint							
pancreas	0.659	0.724	**0.755**	0.751	0.753	0.754	0.734
tumor	0.008	0.235	**0.255**	0.233	0.233	0.221	0.202

TotalSegmentator, TAP-CT: held-out Dice per structure

Structure	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→ LoRA
per-task							
lungs airway	0.903	**0.954**	0.939	–	–	0.951	–
heart	0.848	0.892	0.885	–	–	**0.899**	–
arteries	0.664	**0.781**	0.728	–	–	0.780	–
veins	0.588	**0.739**	0.696	–	–	0.724	–
liver	0.871	0.925	**0.931**	–	–	**0.931**	–
spleen	0.792	0.883	0.890	–	–	**0.913**	–
pancreas	0.644	0.725	0.697	–	–	**0.728**	–
kidneys	0.787	0.850	**0.852**	–	–	0.844	–
gi tract	0.759	0.862	0.842	–	–	**0.864**	–
bladder prostate	0.829	0.844	0.830	–	–	**0.850**	–
glands	0.473	**0.628**	0.516	–	–	0.409	–
brain spinal cord	0.781	0.909	0.841	–	–	**0.914**	–
spine	0.734	0.871	0.869	–	–	**0.879**	–
other bones	0.734	0.882	0.877	–	–	**0.889**	–
muscles	0.800	0.903	0.895	–	–	**0.908**	–
joint							
lungs airway	0.901	0.945	0.822	0.834	0.823	**0.952**	0.943
heart	0.844	0.888	0.887	0.878	0.887	**0.898**	0.884
arteries	0.669	0.790	0.797	0.771	0.804	**0.812**	0.791
veins	0.582	0.754	0.745	0.729	0.743	**0.785**	0.740
liver	0.860	0.925	0.802	0.812	0.801	**0.939**	0.921
spleen	0.772	0.890	0.894	0.907	0.893	**0.927**	0.875
pancreas	0.622	0.710	0.682	0.713	0.694	**0.738**	0.679
kidneys	0.771	0.834	0.839	0.839	0.835	**0.861**	0.831
gi tract	0.757	0.858	0.838	0.854	0.852	**0.875**	0.848
bladder prostate	0.810	0.853	0.840	0.817	0.829	**0.871**	0.841
glands	**0.466**	0.000	0.000	0.000	0.000	0.000	0.000
brain spinal cord	0.785	0.911	0.893	0.893	0.886	**0.918**	0.909
spine	0.732	0.853	0.838	0.850	0.839	**0.864**	0.844
other bones	0.729	0.862	0.859	0.856	0.857	**0.875**	0.854
muscles	0.802	0.895	0.898	0.881	0.895	**0.909**	0.888
joint (8000 steps)							
lungs airway	0.896	0.935	0.882	0.871	0.908	**0.941**	0.932
heart	0.837	0.881	0.861	0.872	0.840	**0.894**	0.880
arteries	0.669	0.781	0.716	0.602	0.650	**0.805**	0.774
veins	0.582	0.720	0.000	0.232	0.075	**0.758**	0.708
liver	0.854	0.916	0.794	0.822	0.855	**0.933**	0.911
spleen	0.738	0.863	0.874	0.888	0.884	**0.916**	0.856
pancreas	0.588	0.662	0.000	0.000	0.000	**0.710**	0.660
kidneys	0.759	0.818	0.564	0.674	0.683	**0.842**	0.815
gi tract	0.755	0.837	0.816	0.812	0.796	**0.857**	0.833
bladder prostate	0.809	0.829	0.000	0.000	0.000	**0.859**	0.816
glands	**0.438**	0.000	0.000	0.000	0.000	0.000	0.000
brain spinal cord	0.786	0.904	0.679	0.599	0.618	**0.913**	0.903
spine	0.727	0.838	0.830	0.833	0.832	**0.851**	0.833
other bones	0.729	0.849	0.845	0.841	0.844	**0.864**	0.845
muscles	0.794	0.885	0.880	0.866	0.874	**0.901**	0.882

SPECTRE segmentation.

AMOS22, SPECTRE: held-out Dice per structure

Structure	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→ LoRA
per-task							
spleen	0.868	0.928	0.932	0.934	**0.938**	0.926	0.933
right kidney	0.704	0.883	**0.911**	0.900	0.902	0.829	0.903
left kidney	0.755	0.874	**0.916**	0.907	0.914	0.780	0.902
gallbladder	0.627	0.729	0.740	0.724	**0.755**	0.728	0.732
esophagus	0.544	0.768	0.764	0.749	0.756	**0.784**	0.772
liver	0.920	0.954	**0.956**	0.952	0.954	0.951	0.955
stomach	0.847	0.908	**0.909**	0.907	0.907	0.898	0.905
aorta	0.782	0.915	**0.918**	0.915	**0.918**	0.916	0.917
postcava	0.653	0.824	**0.839**	0.834	**0.839**	**0.839**	0.828
pancreas	0.701	0.794	**0.813**	0.804	**0.813**	0.801	0.810
right adrenal	0.370	**0.573**	0.519	0.546	0.531	0.532	0.557
left adrenal	0.366	0.572	0.535	0.532	0.538	0.521	**0.576**
duodenum	0.551	0.714	0.743	0.730	**0.747**	0.722	0.725
bladder	0.812	0.880	0.883	0.875	0.883	0.865	**0.885**
prostate uterus	0.738	0.800	0.800	0.784	**0.814**	0.783	0.803
joint							
spleen	0.859	0.911	0.904	0.918	**0.920**	0.901	0.910
right kidney	0.770	0.866	**0.895**	0.876	0.858	0.843	0.876
left kidney	0.763	0.843	0.801	0.869	0.844	**0.878**	0.876
gallbladder	0.647	0.699	0.742	0.716	0.553	**0.753**	0.698
esophagus	0.527	0.000	0.696	0.523	**0.699**	0.000	0.000
liver	0.920	0.948	0.949	0.944	**0.950**	0.948	**0.950**
stomach	0.838	0.885	0.893	0.894	**0.900**	0.874	0.888
aorta	0.772	0.889	0.898	0.885	0.896	**0.906**	0.890
postcava	0.661	0.802	**0.827**	0.818	0.826	0.825	0.807
pancreas	0.704	0.770	0.791	0.776	0.779	**0.792**	0.769
right adrenal	0.369	0.000	0.524	0.500	**0.531**	0.000	0.000
left adrenal	0.351	0.000	0.525	0.434	**0.526**	0.000	0.000
duodenum	0.555	0.678	**0.713**	0.697	0.712	0.708	0.687
bladder	0.796	0.832	**0.856**	0.831	0.847	0.837	0.842
prostate uterus	0.704	0.752	0.745	0.773	**0.785**	0.777	0.764
joint (8000 steps)							
spleen	0.852	0.892	0.888	0.890	**0.907**	0.887	0.886
right kidney	0.759	0.846	0.878	0.870	**0.886**	0.807	0.844
left kidney	0.727	0.832	0.000	0.000	0.000	**0.850**	0.844
gallbladder	0.640	0.686	0.691	0.587	0.686	**0.706**	0.684
esophagus	**0.503**	0.000	0.000	0.000	0.000	0.000	0.000
liver	0.919	0.942	0.924	0.918	0.926	**0.946**	0.942
stomach	0.838	0.872	0.889	0.892	**0.893**	0.865	0.873
aorta	0.759	0.866	0.867	0.733	0.859	**0.889**	0.859
postcava	0.651	0.781	0.700	0.681	0.693	**0.805**	0.781
pancreas	0.695	0.746	0.705	0.722	0.727	**0.757**	0.740
right adrenal	**0.337**	0.000	0.000	0.000	0.000	0.000	0.000
left adrenal	**0.338**	0.000	0.000	0.000	0.000	0.000	0.000
duodenum	0.540	0.628	0.561	0.200	0.540	**0.656**	0.630
bladder	0.786	0.796	0.716	0.713	0.706	**0.839**	0.789
prostate uterus	0.707	0.707	0.705	0.702	0.678	**0.734**	0.531

KiTS23, SPECTRE: held-out Dice per structure

Structure	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→ LoRA
per-task							
kidney	0.903	0.943	**0.948**	0.944	0.947	0.933	0.944
tumor	0.624	0.631	0.697	0.700	**0.718**	0.599	0.659
cyst	0.221	**0.222**	0.194	0.206	0.200	0.051	0.205
joint							
kidney	0.904	0.933	**0.944**	0.943	0.920	0.891	0.935
tumor	0.527	0.634	**0.731**	0.717	0.692	0.557	0.644
cyst	0.161	0.149	0.090	0.103	**0.166**	0.109	0.130

LiTS, SPECTRE: held-out Dice per structure

Structure	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→ LoRA
per-task							
liver	0.945	**0.966**	**0.966**	**0.966**	**0.966**	0.965	**0.966**
tumor	0.431	0.516	**0.546**	0.480	**0.546**	0.512	0.492
joint							
liver	0.936	0.961	0.962	0.962	**0.965**	0.955	0.960
tumor	0.457	0.524	**0.545**	0.511	0.521	0.470	0.526

MSD Pancreas, SPECTRE: held-out Dice per structure

Structure	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→ LoRA
per-task							
pancreas	0.709	0.777	0.790	0.776	**0.791**	0.788	0.780
tumor	0.241	0.255	**0.337**	0.300	0.324	0.245	0.257
joint							
pancreas	0.700	0.757	0.785	0.766	**0.787**	0.773	0.759
tumor	0.218	0.271	**0.315**	0.293	0.314	0.286	0.280

TotalSegmentator, SPECTRE: held-out Dice per structure

Structure	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→ LoRA
per-task							
lungs airway	0.902	**0.953**	0.952	–	–	0.952	–
heart	0.863	0.911	**0.913**	–	–	0.905	–
arteries	0.632	0.787	0.774	–	–	**0.802**	–
veins	0.547	0.738	0.765	–	–	**0.769**	–
liver	0.906	**0.939**	0.930	–	–	0.925	–
spleen	0.870	**0.907**	**0.907**	–	–	0.804	–
pancreas	0.660	**0.748**	0.729	–	–	0.740	–
kidneys	0.787	0.852	**0.856**	–	–	0.803	–
gi tract	0.750	0.854	**0.869**	–	–	0.865	–
bladder prostate	0.808	0.863	0.867	–	–	**0.873**	–
glands	0.408	**0.627**	0.605	–	–	0.565	–
brain spinal cord	0.739	**0.913**	0.902	–	–	0.908	–
spine	0.669	0.858	0.870	–	–	**0.871**	–
other bones	0.678	0.866	0.870	–	–	**0.878**	–
muscles	0.749	0.896	**0.912**	–	–	0.911	–
joint							
lungs airway	0.898	0.929	0.828	0.880	0.854	**0.944**	0.928
heart	0.858	0.891	0.878	0.882	0.877	**0.905**	0.892
arteries	0.630	0.737	0.766	0.751	0.763	**0.783**	0.739
veins	0.540	0.687	0.718	0.716	0.729	**0.753**	0.691
liver	0.905	0.924	0.746	0.826	0.755	**0.930**	0.924
spleen	0.859	0.862	0.649	0.720	0.707	**0.883**	0.867
pancreas	0.644	0.672	0.633	0.636	0.655	**0.701**	0.687
kidneys	0.769	0.812	0.790	0.812	0.794	**0.814**	0.802
gi tract	0.744	0.832	0.854	0.853	0.854	**0.861**	0.833
bladder prostate	0.798	0.795	0.685	0.742	0.597	**0.851**	0.814
glands	**0.379**	0.000	0.000	0.000	0.000	0.000	0.000
brain spinal cord	0.736	0.901	0.903	0.902	0.902	**0.914**	0.899
spine	0.661	0.808	0.825	0.823	0.825	**0.829**	0.806
other bones	0.670	0.823	0.846	0.842	0.847	**0.850**	0.822
muscles	0.736	0.871	0.881	0.891	0.886	**0.901**	0.875
joint (8000 steps)							
lungs airway	**0.896**	0.836	0.878	0.876	0.861	0.860	0.778
heart	0.855	0.877	0.811	0.815	0.812	**0.884**	0.855
arteries	0.619	0.723	0.000	0.000	0.000	**0.768**	0.718
veins	0.521	0.654	0.001	0.000	0.024	**0.706**	0.646
liver	**0.902**	0.817	0.502	0.000	0.256	0.888	0.857
spleen	**0.844**	0.699	0.000	0.000	0.000	0.797	0.469
pancreas	**0.626**	0.000	0.000	0.000	0.000	0.000	0.000
kidneys	0.750	0.740	0.000	0.000	0.000	**0.793**	0.665
gi tract	0.743	0.820	0.799	0.786	0.791	**0.839**	0.818
bladder prostate	**0.797**	0.677	0.000	0.000	0.000	0.772	0.419
glands	**0.345**	0.000	0.000	0.000	0.000	0.000	0.000
brain spinal cord	0.723	0.894	0.881	0.881	0.871	**0.905**	0.892
spine	0.657	0.790	0.794	0.778	0.794	**0.811**	0.784
other bones	0.669	0.810	0.831	0.819	0.829	**0.836**	0.811
muscles	0.731	0.854	0.884	0.853	0.883	**0.886**	0.843

VoCo segmentation.

AMOS22, VoCo: held-out Dice per structure

Structure	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→ LoRA
per-task							
spleen	0.000	0.905	0.899	0.884	0.893	**0.917**	0.898
right kidney	0.062	0.685	0.648	0.680	0.655	**0.701**	0.688
left kidney	0.000	0.795	0.751	0.762	0.764	**0.893**	0.766
gallbladder	0.000	0.751	0.691	0.699	0.695	**0.752**	0.740
esophagus	0.000	0.790	**0.812**	0.807	0.810	0.801	0.788
liver	0.544	0.942	0.938	0.935	0.936	**0.947**	0.943
stomach	0.011	0.858	0.833	0.830	0.832	**0.865**	0.856
aorta	0.003	0.930	0.931	**0.932**	0.930	**0.932**	0.927
postcava	0.000	0.858	**0.864**	0.863	0.863	0.862	0.856
pancreas	0.000	0.817	0.817	0.813	0.816	**0.826**	0.817
right adrenal	0.000	**0.676**	0.646	0.651	0.650	0.670	0.671
left adrenal	0.000	0.694	0.672	0.671	0.680	**0.702**	0.685
duodenum	0.000	0.733	0.700	0.714	0.704	**0.743**	0.723
bladder	0.000	0.861	0.811	0.802	0.800	**0.867**	0.860
prostate uterus	0.000	0.776	0.717	0.697	0.716	**0.787**	0.761
joint							
spleen	0.000	0.890	0.848	0.839	0.866	**0.900**	0.881
right kidney	0.000	0.870	0.889	0.861	0.876	**0.910**	0.875
left kidney	0.000	0.897	0.870	0.877	0.867	**0.914**	0.892
gallbladder	0.000	0.698	0.692	0.698	0.695	**0.722**	0.689
esophagus	0.000	0.733	0.724	0.725	0.717	**0.745**	0.730
liver	0.570	0.944	0.941	0.940	0.941	**0.947**	0.943
stomach	0.000	0.838	0.819	0.818	0.826	**0.855**	0.836
aorta	0.000	0.893	0.883	0.885	0.888	**0.896**	0.891
postcava	0.000	0.820	0.830	0.826	0.824	**0.832**	0.817
pancreas	0.000	0.788	0.797	0.791	0.795	**0.799**	0.786
right adrenal	0.000	0.631	0.632	0.621	0.626	**0.648**	0.623
left adrenal	0.000	0.648	0.640	0.638	0.638	**0.657**	0.639
duodenum	0.000	0.677	0.684	0.675	0.672	**0.692**	0.672
bladder	0.000	0.764	0.715	0.708	0.712	**0.769**	0.761
prostate uterus	0.000	0.000	0.467	0.459	**0.514**	0.000	0.000
joint (8000 steps)							
spleen	0.000	0.826	**0.871**	0.866	0.863	0.859	0.817
right kidney	0.000	0.854	**0.889**	0.855	0.885	0.880	0.838
left kidney	0.000	0.866	0.878	0.857	0.873	**0.891**	0.855
gallbladder	0.000	0.648	**0.725**	0.705	0.720	0.688	0.656
esophagus	0.000	0.712	0.687	0.673	0.693	**0.713**	0.701
liver	0.577	0.934	0.938	0.927	0.933	**0.940**	0.933
stomach	0.000	0.806	0.822	0.819	0.824	**0.831**	0.814
aorta	0.000	0.875	0.884	0.883	**0.888**	0.882	0.868
postcava	0.000	0.793	**0.813**	0.802	0.811	0.802	0.788
pancreas	0.000	0.770	0.777	0.761	**0.781**	0.776	0.762
right adrenal	0.000	0.592	0.524	0.534	0.532	**0.610**	0.587
left adrenal	0.000	0.601	0.000	0.000	0.000	**0.617**	0.592
duodenum	0.000	0.635	0.604	0.596	0.630	**0.654**	0.624
bladder	0.000	0.727	**0.741**	0.737	0.735	0.740	0.720
prostate uterus	**0.000**	**0.000**	**0.000**	**0.000**	**0.000**	**0.000**	**0.000**

KiTS23, VoCo: held-out Dice per structure

Structure	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→ LoRA
per-task							
kidney	0.052	0.924	0.904	0.898	0.903	**0.934**	0.923
tumor	0.000	**0.574**	0.416	0.407	0.386	**0.574**	0.518
cyst	0.000	0.060	0.183	**0.194**	0.191	0.184	0.000
joint							
kidney	0.111	0.903	0.868	0.888	0.875	**0.906**	0.895
tumor	0.000	0.490	0.478	0.504	0.485	**0.539**	0.462
cyst	0.000	0.000	0.157	0.157	**0.160**	0.000	0.000

LiTS, VoCo: held-out Dice per structure

Structure	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→ LoRA
per-task							
liver	0.683	0.956	0.955	0.951	0.953	**0.959**	0.955
tumor	0.000	0.511	0.429	0.437	0.423	**0.538**	0.485
joint							
liver	0.691	0.938	0.946	0.944	0.945	**0.947**	0.939
tumor	0.000	**0.526**	0.494	0.477	0.491	**0.526**	0.513

MSD Pancreas, VoCo: held-out Dice per structure

Structure	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→ LoRA
per-task							
pancreas	0.000	0.797	0.788	0.785	0.785	**0.802**	0.794
tumor	0.000	**0.341**	0.291	0.305	0.296	0.330	0.323
joint							
pancreas	0.000	0.781	0.778	0.774	0.779	**0.784**	0.781
tumor	0.000	0.337	0.325	0.317	0.325	0.335	**0.341**

TotalSegmentator, VoCo: held-out Dice per structure

Structure	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→ LoRA
per-task							
lungs airway	0.161	**0.932**	0.820	–	–	**0.932**	–
heart	0.030	0.858	0.798	–	–	**0.878**	–
arteries	0.000	0.790	0.751	–	–	**0.801**	–
veins	0.000	0.748	0.743	–	–	**0.758**	–
liver	0.552	0.885	0.849	–	–	**0.894**	–
spleen	0.000	0.738	0.627	–	–	**0.820**	–
pancreas	0.000	0.703	0.654	–	–	**0.720**	–
kidneys	0.050	0.791	0.725	–	–	**0.804**	–
gi tract	0.013	0.832	0.814	–	–	**0.844**	–
bladder prostate	0.000	0.815	0.605	–	–	**0.830**	–
glands	0.000	0.674	0.624	–	–	**0.687**	–
brain spinal cord	0.014	0.932	0.931	–	–	**0.938**	–
spine	0.003	0.884	**0.897**	–	–	0.891	–
other bones	0.000	0.891	0.883	–	–	**0.893**	–
muscles	0.003	0.862	0.848	–	–	**0.876**	–
joint							
lungs airway	0.226	0.880	0.808	0.802	0.791	**0.902**	0.812
heart	0.000	0.812	0.807	0.781	0.798	**0.832**	0.812
arteries	0.000	0.742	0.750	0.743	0.746	**0.760**	0.739
veins	0.000	0.675	0.697	0.691	0.695	**0.710**	0.668
liver	0.471	0.884	0.842	0.819	0.850	**0.897**	0.886
spleen	0.000	0.709	0.592	0.559	0.594	**0.718**	0.673
pancreas	0.000	0.620	0.629	0.598	0.606	**0.656**	0.614
kidneys	0.000	0.740	0.705	0.691	0.729	**0.776**	0.742
gi tract	0.000	0.802	0.815	0.799	0.812	**0.816**	0.797
bladder prostate	0.000	0.678	0.471	0.517	0.559	**0.698**	0.659
glands	**0.000**	**0.000**	**0.000**	**0.000**	**0.000**	**0.000**	**0.000**
brain spinal cord	0.000	0.926	0.923	0.923	0.924	**0.931**	0.925
spine	0.010	0.849	0.850	0.848	0.849	**0.854**	0.847
other bones	0.000	0.847	0.859	0.855	0.858	**0.861**	0.847
muscles	0.000	0.819	0.825	0.828	0.821	**0.839**	0.819
joint (8000 steps)							
lungs airway	0.170	**0.802**	0.789	0.756	0.793	0.792	0.795
heart	0.000	0.775	0.730	0.736	0.727	**0.784**	0.776
arteries	0.000	0.689	0.673	0.665	0.659	**0.711**	0.685
veins	0.000	0.629	0.620	0.582	0.611	**0.651**	0.621
liver	0.478	0.848	0.805	0.781	0.791	**0.880**	0.792
spleen	0.000	0.547	0.597	0.544	0.570	**0.611**	0.442
pancreas	0.000	0.544	0.557	0.521	0.536	**0.595**	0.550
kidneys	0.000	0.699	0.690	0.669	0.675	**0.726**	0.688
gi tract	0.000	0.775	0.757	0.738	0.757	**0.794**	0.774
bladder prostate	0.000	0.427	0.371	0.419	0.295	**0.623**	0.010
glands	**0.000**	**0.000**	**0.000**	**0.000**	**0.000**	**0.000**	**0.000**
brain spinal cord	0.000	0.917	0.894	0.892	0.892	**0.921**	0.918
spine	0.000	0.830	0.838	0.833	**0.841**	0.834	0.823
other bones	0.000	0.837	**0.854**	0.845	0.852	0.848	0.839
muscles	0.000	0.775	**0.810**	0.762	0.796	0.789	0.778